

External Memory Interface (EMIF)

This chapter describes the external memory Interface (EMIF).

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17.1 Introduction

17.1.1 Purpose of the Peripheral

This EMIF memory controller is compliant with the JESD21-C SDR SDRAM memories utilizing a 16-bit data bus. The purpose of this EMIF is to provide a means for the CPU to connect to a variety of external devices including:

- Single data rate (SDR) SDRAM
- Asynchronous devices including NOR Flash and SRAM

The most common use for the EMIF is to interface with both a flash device and an SDRAM device simultaneously. [Section 17.4](#) contains an example of operating the EMIF in this configuration.

17.1.2 Features

The EMIF includes many features to enhance the ease and flexibility of connecting to external SDR SDRAM and asynchronous devices.

17.1.2.1 Asynchronous Memory Support

EMIF supports asynchronous:

- SRAM memories
- NOR Flash memories

The EMIF data bus width is up to 16 bits and there are up to 13 address lines. There is an external wait input that allows slower asynchronous memories to extend the memory access. The EMIF module supports up to 3 chip selects (EMIF_nCS[4:2]). Each chip select has the following individually programmable attributes:

- Data Bus Width
- Read cycle timings: setup, hold, strobe
- Write cycle timings: setup, hold, strobe
- Bus turn-around time
- Extended Wait Option with Programmable Timeout
- Select Strobe option

17.1.2.2 Synchronous DRAM Memory Support

The EMIF module supports 16-bit SDRAM in addition to the asynchronous memories listed in [Section 17.1.2.1](#). It has a single SDRAM chip select (EMIF_nCS[0]). SDRAM configurations that are supported are:

- One, Two, and Four Bank SDRAM devices
- Devices with Eight, Nine, Ten, and Eleven Column Address
- CAS Latency of two or three clock cycles
- 16-bit Data Bus Width
- 3.3V LVCMOS Interface

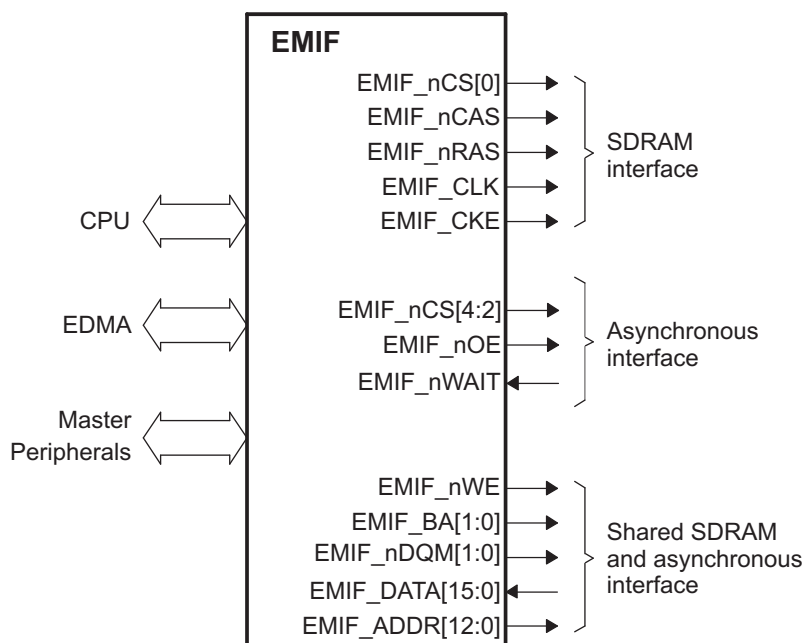
Additionally, the EMIF supports placing the SDRAM in Self-Refresh and Powerdown modes. Self-refresh mode allows the SDRAM to be put in a low-power state while still retaining memory contents; since the SDRAM will continue to refresh itself even without clocks from the microcontroller. Powerdown mode achieves even lower power, except the microcontroller must periodically wake up and issue refreshes if data retention is required.

Note that the EMIF module does not support Mobile SDRAM devices.

17.1.3 Functional Block Diagram

Figure 17-1 illustrates the connections between the EMIF and its internal requesters, along with the external EMIF pins. Section 17.2.2 contains a description of the entities internal to the SoC that can send requests to the EMIF, along with their prioritization. Section 17.2.3 describes the EMIF external pins and summarizes their purpose when interfacing with SDRAM and asynchronous devices.

Figure 17-1. EMIF Functional Block Diagram



17.2 EMIF Module Architecture

This section provides details about the architecture and operation of the EMIF. Both, SDRAM and asynchronous Interface are covered, along with other system-related issues such as clock control.

17.2.1 EMIF Clock Control

The EMIF clock is output on the EMIF_CLK pin and should be used when interfacing to external SDRAM devices. The EMIF module gets the VCLK3 clock domain as the input. This clock domain is running at half the frequency of the main oscillator by default, that is, between 2.5MHz to 10MHz. The VCLK3 frequency is divided down from the HCLK domain frequency by a programmable divider (/1 to /16). Refer the Architecture chapter of the device technical reference manual for more information on configuring the VCLK3 domain frequency.

17.2.2 EMIF Requests

Different sources within the SoC can make requests to the EMIF. These requests consist of accesses to SDRAM memory, asynchronous memory, and EMIF registers. The EMIF can process only one request at a time. Therefore a high performance crossbar switch exists within the SoC to provide prioritized requests from the different sources to the EMIF. The sources are:

1. CPU
2. DMA
3. Other master peripherals

If a request is submitted from two or more sources simultaneously, the crossbar switch will forward the highest priority request to the EMIF first. Upon completion of a request, the crossbar switch again evaluates the pending requests and forwards the highest priority pending request to the EMIF.

When the EMIF receives a request, it may or may not be immediately processed. In some cases, the EMIF will perform one or more auto refresh cycles before processing the request. For details on the EMIF's internal arbitration between performing requests and performing auto refresh cycles, see [Section 17.2.13](#).

17.2.3 EMIF Signal Descriptions

This section describes the function of each of the EMIF signals.

Table 17-1. EMIF Pins Used to Access Both SDRAM and Asynchronous Memories

Pins(s)	I/O	Description
EMIF_DATA[15:0]	I/O	EMIF data bus.
EMIF_ADDR[12:0]	O	EMIF address bus. When interfacing to an SDRAM device, these pins are primarily used to provide the row and column address to the SDRAM. The mapping from the internal program address to the external values placed on these pins can be found in Table 17-13 . EMIF_A[10] is also used during the PRE command to select which banks to deactivate. When interfacing to an asynchronous device, these pins are used in conjunction with the EMIF_BA pins to form the address that is sent to the device. The mapping from the internal program address to the external values placed on these pins can be found in Section 17.2.6.1 .
EMIF_BA[1:0]	O	EMIF bank address. When interfacing to an SDRAM device, these pins are used to provide the bank address inputs to the SDRAM. The mapping from the internal program address to the external values placed on these pins can be found in Table 17-13 . When interfacing to an asynchronous device, these pins are used in conjunction with the EMIF_A pins to form the address that is sent to the device. The mapping from the internal program address to the external values placed on these pins can be found in Section 17.2.6.1 .
EMIF_nDQM[1:0]	O	Active-low byte enables. When interfacing to SDRAM, these pins are connected to the DQM pins of the SDRAM to individually enable/disable each of the bytes in a data access. When interfacing to an asynchronous device, these pins are connected to byte enables. See Section 17.2.6 for details.

Table 17-1. EMIF Pins Used to Access Both SDRAM and Asynchronous Memories (continued)

Pin(s)	I/O	Description
EMIF_nWE	O	Active-low write enable. When interfacing to SDRAM, this pin is connected to the nWE pin of the SDRAM and is used to send commands to the device. When interfacing to an asynchronous device, this pin provides a signal which is active-low during the strobe period of an asynchronous write access cycle.

Table 17-2. EMIF Pins Specific to SDRAM

Pin(s)	I/O	Description
EMIF_nCS[0]	O	Active-low chip enable pin for SDRAM devices. This pin is connected to the chip-select pin of the attached SDRAM device and is used for enabling/disabling commands. By default, the EMIF keeps this SDRAM chip select active, even if the EMIF is not interfaced with an SDRAM device. This pin is deactivated when accessing the asynchronous memory bank and is reactivated on completion of the asynchronous access.
EMIF_nRAS	O	Active-low row address strobe pin. This pin is connected to the nRAS pin of the attached SDRAM device and is used for sending commands to the device.
EMIF_nCAS	O	Active-low column address strobe pin. This pin is connected to the nCAS pin of the attached SDRAM device and is used for sending commands to the device.
EMIF_CKE	O	Clock enable pin. This pin is connected to the CKE pin of the attached SDRAM device and is used for issuing the SELF REFRESH command which places the device in self refresh mode. See Section 17.2.5.7 for details.
EMIF_CLK	O	SDRAM clock pin. This pin is connected to the CLK pin of the attached SDRAM device. See Section 17.2.1 for details on the clock signal.

Table 17-3. EMIF Pins Specific to Asynchronous Memory

Pin(s)	I/O	Description
EMIF_nCS[4:2]	O	Active-low chip enable pins for asynchronous devices. These pins are meant to be connected to the chip-select pins of the attached asynchronous device. These pins are active only during accesses to the asynchronous memory.
EMIF_nWAIT	I	Wait input with programmable polarity. A connected asynchronous device can extend the strobe period of an access cycle by asserting the EMIF_nWAIT input to the EMIF as described in Section 17.2.6.6 . To enable this functionality, the EW bit in the asynchronous 1 configuration register (CE2CFG) must be set to 1. In addition, the WP0 bit in CE2CFG must be configured to define the polarity of the EMIF_nWAIT pin.
EMIF_nOE	O	Active-low pin enable for asynchronous devices. This pin provides a signal which is active-low during the strobe period of an asynchronous read access cycle.

17.2.4 EMIF Signal Multiplexing Control

Several EMIF signals are multiplexed with other functions on this microcontroller. Please refer to the I/O Multiplexing Module chapter of the technical reference manual for more information on how to enable the output of these EMIF signals.

17.2.5 SDRAM Controller and Interface

The EMIF can gluelessly interface to most standard SDR SDRAM devices and supports such features as self refresh mode and prioritized refresh. In addition, it provides flexibility through programmable parameters such as the refresh rate, CAS latency, and many SDRAM timing parameters. The following sections include details on how to Interface and properly configure the EMIF to perform read and write operations to externally connected SDR SDRAM devices. Also, [Section 17.4](#) provides a detailed example of interfacing the EMIF to a common SDRAM device.

17.2.5.1 SDRAM Commands

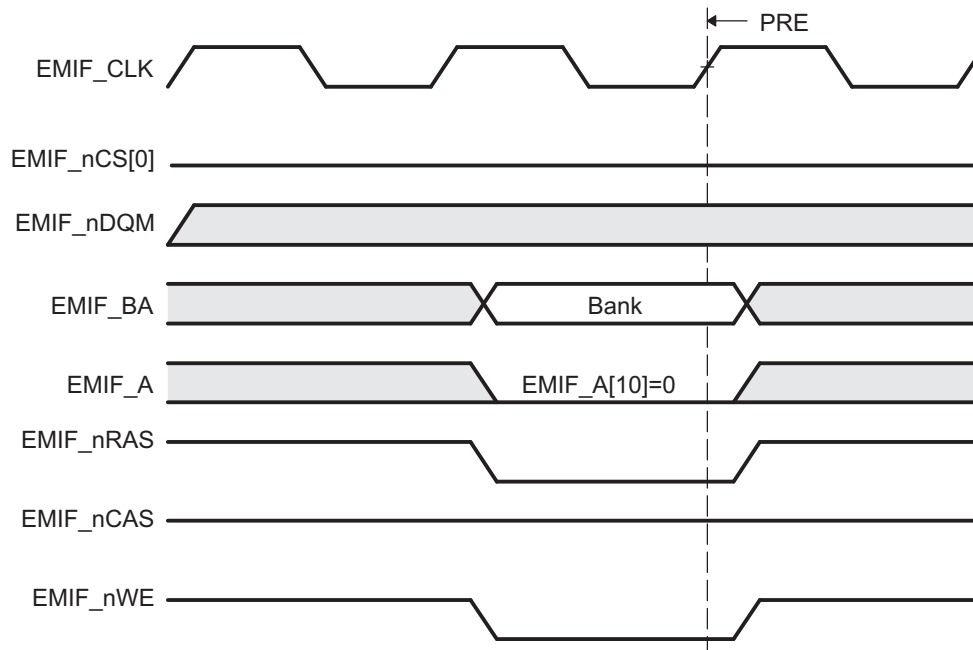
The EMIF supports the SDRAM commands described in [Table 17-4](#). [Table 17-5](#) shows the truth table for the SDRAM commands, and an example timing waveform of the PRE command is shown in [Figure 17-2](#). EMIF_A[10] is pulled low in this example to deactivate only the bank specified by the EMIF_BA pins.

Table 17-4. EMIF SDRAM Commands

Command	Function
PRE	Precharge. Depending on the value of EMIF_A[10], the PRE command either deactivates the open row in all banks (EMIF_A[10] = 1) or only the bank specified by the EMIF_BA[1:0] pins (EMIF_A[10] = 0).
ACTV	Activate. The ACTV command activates the selected row in a particular bank for the current access.
READ	Read. The READ command outputs the starting column address and signals the SDRAM to begin the burst read operation. Address EMIF_A[10] is always pulled low to avoid auto precharge. This allows for better bank interleaving performance.
WRT	Write. The WRT command outputs the starting column address and signals the SDRAM to begin the burst write operation. Address EMIF_A[10] is always pulled low to avoid auto precharge. This allows for better bank interleaving performance.
BT	Burst terminate. The BT command is used to truncate the current read or write burst request.
LMR	Load mode register. The LMR command sets the mode register of the attached SDRAM devices and is only issued during the SDRAM initialization sequence described in Section 17.2.5.4 .
REFR	Auto refresh. The REFR command signals the SDRAM to perform an auto refresh according to its internal address.
SLFR	Self refresh. The self refresh command places the SDRAM into self refresh mode, during which it provides its own clock signal and auto refresh cycles.
NOP	No operation. The NOP command is issued during all cycles in which one of the above commands is not issued.

Table 17-5. Truth Table for SDRAM Commands

SDRAM Pins:	CKE	nCS	nRAS	nCAS	nWE	BA[1:0]	A[12:11]	A[10]	A[9:0]
EMIF Pins:	EMIF_CKE	EMIF_nCS[0]	EMIF_nRAS	EMIF_nCAS	EMIF_nWE	EMIF_BA[1:0]	EMIF_A[12:11]	EMIF_A[10]	EMIF_A[9:0]
PRE	H	L	L	H	L	Bank/X	X	L/H	X
ACTV	H	L	L	H	H	Bank	Row	Row	Row
READ	H	L	H	L	H	Bank	Column	L	Column
WRT	H	L	H	L	L	Bank	Column	L	Column
BT	H	L	H	H	L	X	X	X	X
LMR	H	L	L	L	L	X	Mode	Mode	Mode
REFR	H	L	L	L	H	X	X	X	X
SLFR	L	L	L	L	H	X	X	X	X
NOP	H	L	H	H	H	X	X	X	X

Figure 17-2. Timing Waveform of SDRAM PRE Command


17.2.5.2 Interfacing to SDRAM

The EMIF supports a glueless interface to SDRAM devices with the following characteristics:

- Pre-charge bit is A[10]
- The number of column address bits is 8, 9, 10, or 11.
- The number of row address bits is 13, 14, 15, or 16.
- The number of internal banks is 1, 2, or 4.

Figure 17-3 shows an interface between the EMIF and a 2M × 16 × 4 bank SDRAM device, and Figure 17-4 shows an interface between the EMIF and a 512K × 16 × 2 bank SDRAM device. For devices supporting 16-bit interface, refer to Table 17-6 for list of commonly-supported SDRAM devices and the required connections for the address pins.

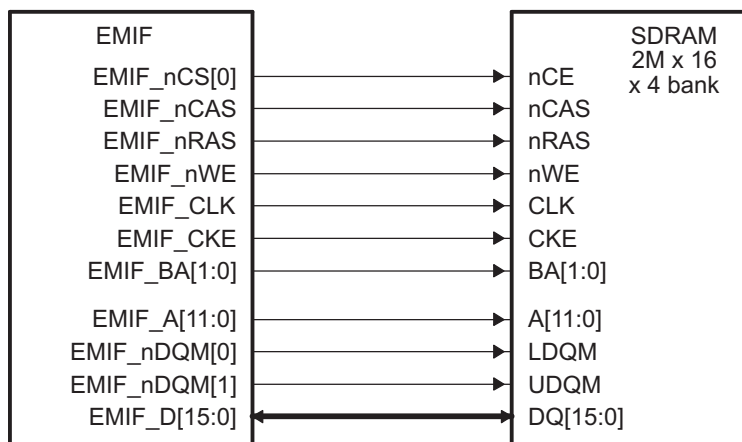
Figure 17-3. EMIF to 2M × 16 × 4 bank SDRAM Interface


Figure 17-4. EMIF to 512K × 16 × 2 bank SDRAM Interface

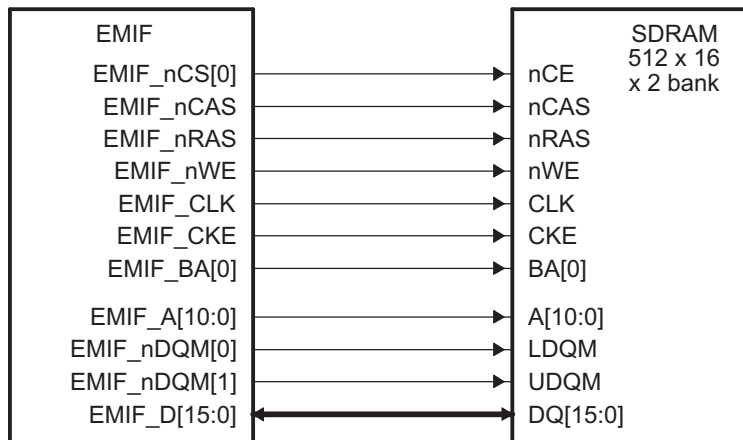


Table 17-6. 16-bit EMIF Address Pin Connections

SDRAM Size	Width	Banks	Device	Address Pins
16M bits	×16	2	SDRAM	A[10:0]
			EMIF	EMIF_A[10:0]
64M bits	×16	4	SDRAM	A[11:0]
			EMIF	EMIF_A[11:0]
128M bits	×16	4	SDRAM	A[11:0]
			EMIF	EMIF_A[11:0]
256M bits	×16	4	SDRAM	A[12:0]
			EMIF	EMIF_A[12:0]
512M bits	×16	4	SDRAM	A[12:0]
			EMIF	EMIF_A[12:0]

17.2.5.3 SDRAM Configuration Registers

The operation of the EMIF's SDRAM interface is controlled by programming the appropriate configuration registers. This section describes the purpose and function of each configuration register, but [Section 17.3](#) should be referred for a more detailed description of each register, including the default registers values and bit-field positions. The following tables list the four such configuration registers, along with a description of each of their programmable fields.

NOTE: Writing to any of the fields: NM, CL, IBANK, and PAGESIZE in the SDRAM configuration register (SDCR) causes the EMIF to abandon whatever it is currently doing and trigger the SDRAM initialization procedure described in [Section 17.2.5.4](#).

Table 17-7. Description of the SDRAM Configuration Register (SDCR)

Parameter	Description
SR	This bit controls entering and exiting of the Self-Refresh mode. The field should be written using a byte-write to the upper byte of SDCR to avoid triggering the SDRAM initialization sequence.
PD	This bit controls entering and exiting of the Power down mode. The field should be written using a byte-write to the upper byte of SDCR to avoid triggering the SDRAM initialization sequence. If both SR and PD bits are set, the EMIF will go into Self Refresh.
PDWR	Perform refreshes during Power Down. Writing a 1 to this bit will cause the EMIF to exit the power down state and issue an AUTO REFRESH command every time Refresh May level is set. The field should be written using a byte-write to the upper byte of SDCR to avoid triggering the SDRAM initialization sequence. This bit should be set along with PD when entering power-down mode.
NM	Narrow Mode. This bit defines the width of the data bus between the EMIF and the attached SDRAM device. When set to 1, the data bus is set to 16-bits. When set to 0, the data bus is set to 32-bits. This bit must always be set to 1.
CL	CAS latency. This field defines the number of clock cycles between when an SDRAM issues a READ command and when the first piece of data appears on the bus. The value in this field is sent to the attached SDRAM device via the LOAD MODE REGISTER command during the SDRAM initialization procedure as described in Section 17.2.5.4 . Only, values of 2h (CAS latency = 2) and 3h (CAS latency = 3) are supported and should be written to this field. A 1 must be simultaneously written to the BIT11_9LOCK bit field of SDCR in order to write to the CL bit field.
IBANK	Number of Internal SDRAM Banks. This field defines the number of banks inside the attached SDRAM devices in the following way: - When IBANK = 0, 1 internal bank is used - When IBANK = 1h, 2 internal banks are used - When IBANK = 2h, 4 internal banks are used This field value affects the mapping of logical addresses to SDRAM row, column, and bank addresses. See Section 17.2.5.11 for details.
PAGESIZE	Page Size. This field defines the internal page size of the attached SDRAM devices in the following way: - When PAGESIZE = 0, 256-word pages are used - When PAGESIZE = 1h, 512-word pages are used - When PAGESIZE = 2h, 1024-word pages are used - When PAGESIZE = 3h, 2048-word pages are used This field value affects the mapping of logical addresses to SDRAM row, column, and bank addresses. See Section 17.2.5.11 for details.

Table 17-8. Description of the SDRAM Refresh Control Register (SDRCR)

Parameter	Description
RR	Refresh Rate. This field controls the rate at which attached SDRAM devices will be refreshed. The following equation can be used to determine the required value of RR for an SDRAM device: $RR = f_{EMIF_CLK} / (\text{Required SDRAM Refresh Rate})$ More information about the operation of the SDRAM refresh controller can be found in Section 17.2.5.6 .

Table 17-9. Description of the SDRAM Timing Register (SDTIMR)

Parameter	Description
T_RFC	SDRAM Timing Parameters. These fields configure the EMIF to comply with the AC timing requirements of the attached SDRAM devices. This allows the EMIF to avoid violating SDRAM timing constraints and to more efficiently schedule its operations. More details about each of these parameters can be found in the register description in Section 17.3.6 . These parameters should be set to satisfy the corresponding timing requirements found in the SDRAM's datasheet.
T_RP	
T_RCD	
T_WR	
T_RAS	
T_RC	
T_RRD	

Table 17-10. Description of the SDRAM Self Refresh Exit Timing Register (SDSRETR)

Parameter	Description
T_XS	Self Refresh Exit Parameter. The T_XS field of this register informs the EMIF about the minimum number of EMIF_CLK cycles required between exiting Self Refresh and issuing any command. This parameter should be set to satisfy the t_{XSR} value for the attached SDRAM device.

17.2.5.4 SDRAM Auto-Initialization Sequence

The EMIF automatically performs an SDRAM initialization sequence, regardless of whether it is interfaced to an SDRAM device, when either of the following two events occur:

- The EMIF comes out of reset. No memory accesses to the SDRAM and Asynchronous interfaces are performed until this auto-initialization is complete.
- A write is performed to any of the three least significant bytes of the SDRAM configuration register (SDCR)

An SDRAM initialization sequence consists of the following steps:

1. If the initialization sequence is activated by a write to SDCR, and if any of the SDRAM banks are open, the EMIF issues a PRE command with EMIF_A[10] held high to indicate all banks. This is done so that the maximum ACTV to PRE timing for an SDRAM is not violated.
2. The EMIF drives EMIF_CKE high and begins continuously issuing NOP commands until eight SDRAM refresh intervals have elapsed. An SDRAM refresh interval is equal to the value of the RR field of SDRAM refresh control register (SDRCR), divided by the frequency of EMIF_CLK (RR/f_{EMIF_CLK}). This step is used to avoid violating the Power-up constraint of most SDRAM devices that requires 200 μ s (sometimes 100 μ s) between receiving stable Vdd and CLK and the issuing of a PRE command. Depending on the frequency of EMIF_CLK, this step may or may not be sufficient to avoid violating the SDRAM constraint. See [Section 17.2.5.5](#) for more information.
3. After the refresh intervals have elapsed, the EMIF issues a PRE command with EMIF_A[10] held high to indicate all banks.
4. The EMIF issues eight AUTO REFRESH commands.
5. The EMIF issues the LMR command with the EMIF_A[9:0] pins set as described in [Table 17-11](#).
6. Finally, the EMIF performs a refresh cycle, which consists of the following steps:
 - a. Issuing a PRE command with EMIF_A[10] held high if any banks are open
 - b. Issuing an REF command

Table 17-11. SDRAM LOAD MODE REGISTER Command

EMIF_A[9:7]	EMIF_A[6:4]	EMIF_A[3]	EMIF_A[2:0]
0 (Write bursts are of the programmed burst length in EMIF_A[2:0])	These bits control the CAS latency of the SDRAM and are set according to CL field in the SDRAM configuration register (SDCR) as follows: - If CL = 2, EMIF_A[6:4] = 2h (CAS latency = 2) - If CL = 3, EMIF_A[6:4] = 3h (CAS latency = 3)	0 (Sequential Burst Type. Interleaved Burst Type not supported)	These bits control the burst length of the SDRAM and are set according to the NM field in the SDRAM configuration register (SDCR) as follows: - If NM = 0, EMIF_A[2:0] = 2h (Burst Length = 4) - If NM = 1, EMIF_A[2:0] = 3h (Burst Length = 8)

17.2.5.5 SDRAM Configuration Procedure

There are two different SDRAM configuration procedures. Although EMIF automatically performs the SDRAM initialization sequence described in [Section 17.2.5.4](#) when coming out of reset, it is recommended to follow one of the procedures listed below before performing any EMIF memory requests. Procedure A should be followed if it is determined that the SDRAM Power-up constraint was not violated during the SDRAM Auto-Initialization Sequence detailed in [Section 17.2.5.4](#) on coming out of Reset. The SDRAM Power-up constraint specifies that 200 μ s (sometimes 100 μ s) should exist between receiving stable Vdd and CLK and the issuing of a PRE command. Procedure B should be followed if the SDRAM Power-up constraint was violated. The 200 μ s (100 μ s) SDRAM Power-up constraint will be violated if the frequency of EMIF_CLK is greater than 50 MHz (100 MHz for 100 μ s SDRAM power-up constraint) during SDRAM Auto-Initialization Sequence. Procedure B should be followed if there is any doubt that the Power-up constraint was not met.

Procedure A — Following is the procedure to be followed if the SDRAM Power-up constraint was NOT violated:

1. Place the SDRAM into Self-Refresh Mode by setting the SR bit of SDCR to 1. A byte-write to the upper byte of SDCR should be used to avoid restarting the SDRAM Auto-Initialization Sequence described in [Section 17.2.5.4](#). The SDRAM should be placed into Self-Refresh mode when changing the frequency of EMIF_CLK to avoid incurring the 200 μ s Power-up constraint again.
2. Configure the desired EMIF_CLK clock frequency. The frequency of the memory clock must meet the timing requirements in the SDRAM manufacturer's documentation and the timing limitations shown in the electrical specifications of the device datasheet.
3. Remove the SDRAM from Self-Refresh Mode by clearing the SR bit of SDCR to 0. A byte-write to the upper byte of SDCR should be used to avoid restarting the SDRAM Auto-Initialization Sequence described in [Section 17.2.5.4](#).
4. Program SDTIMR and SDSRETR to satisfy the timing requirements for the attached SDRAM device. The timing parameters should be taken from the SDRAM datasheet.
5. Program the RR field of SDRCR to match that of the attached device's refresh interval. See [Section 17.2.5.6.1](#) details on determining the appropriate value.
6. Program SDCR to match the characteristics of the attached SDRAM device. This will cause the auto-initialization sequence in [Section 17.2.5.4](#) to be re-run. This second initialization generally takes much less time due to the increased frequency of EMIF_CLK.

Procedure B — Following is the procedure to be followed if the SDRAM Power-up constraint was violated:

1. Configure the desired EMIF_CLK clock frequency. The frequency of the memory clock must meet the timing requirements in the SDRAM manufacturer's documentation and the timing limitations shown in the electrical specifications of the device datasheet.
2. Program SDTIMR and SDSRETR to satisfy the timing requirements for the attached SDRAM device. The timing parameters should be taken from the SDRAM datasheet.
3. Program the RR field of SDRCR such that the following equation is satisfied: $(RR \times 8)/(f_{EMIF_CLK}) > 200 \mu s$ (sometimes 100 μ s). For example, an EMIF_CLK frequency of 100 MHz would require setting RR to 2501 (9C5h) or higher to meet a 200 μ s constraint.

4. Program SDCR to match the characteristics of the attached SDRAM device. This will cause the auto-initialization sequence in [Section 17.2.5.4](#) to be re-run with the new value of RR.
5. Perform a read from the SDRAM to assure that step 5 of this procedure will occur after the initialization process has completed. Alternatively, wait for 200 μ s instead of performing a read.
6. Finally, program the RR field to match that of the attached device's refresh interval. See [Section 17.2.5.6.1](#) details on determining the appropriate value.

After following the above procedure, the EMIF is ready to perform accesses to the attached SDRAM device. See [Section 17.4](#) for an example of configuring the SDRAM interface.

17.2.5.6 EMIF Refresh Controller

An SDRAM device requires that each of its rows be refreshed at a minimum required rate. The EMIF can meet this constraint by performing auto refresh cycles at or above this required rate. An auto refresh cycle consists of issuing a PRE command to all banks of the SDRAM device followed by issuing a REFR command. To inform the EMIF of the required rate for performing auto refresh cycles, the RR field of the SDRAM refresh control register (SDRCR) must be programmed. The EMIF will use this value along with two internal counters to automatically perform auto refresh cycles at the required rate. The auto refresh cycles cannot be disabled, even if the EMIF is not interfaced with an SDRAM. The remainder of this section details the EMIF's refresh scheme and provides an example for determining the appropriate value to place in the RR field of SDRCR.

The two counters used to perform auto-refresh cycles are a 13-bit refresh interval counter and a 4-bit refresh backlog counter. At reset and upon writing to the RR field, the refresh interval counter is loaded with the value from RR field and begins decrementing, by one, each EMIF clock cycle. When the refresh interval counter reaches zero, the following actions occur:

- The refresh interval counter is reloaded with the value from the RR field and restarts decrementing.
- The 4-bit refresh backlog counter increments unless it has already reached its maximum value.

The refresh backlog counter records the number of auto refresh cycles that the EMIF currently has outstanding. This counter is decremented by one each time an auto refresh cycle is performed and incremented by one each time the refresh interval counter expires. The refresh backlog counter saturates at the values of 0000b and 1111b. The EMIF uses the refresh backlog counter to determine the urgency with which an auto refresh cycle should be performed. The four levels of urgency are described in [Table 17-12](#). This refresh scheme allows the required refreshes to be performed with minimal impact on access requests.

Table 17-12. Refresh Urgency Levels

Urgency Level	Refresh Backlog Counter Range	Action Taken
Refresh May	1-3	An auto-refresh cycle is performed only if the EMIF has no requests pending and none of the SDRAM banks are open.
Refresh Release	4-7	An auto-refresh cycle is performed if the EMIF has no requests pending, regardless of whether any SDRAM banks are open.
Refresh Need	8-11	An auto-refresh cycle is performed at the completion of the current access unless there are read requests pending.
Refresh Must	12-15	Multiple auto-refresh cycles are performed at the completion of the current access until the Refresh Release urgency level is reached. At that point, the EMIF can begin servicing any new read or write requests.

17.2.5.6.1 Determining the Appropriate Value for the RR Field

The value that should be programmed into the RR field of SDCR can be calculated by using the frequency of the EMIF_CLK signal ($f_{\text{EMIF_CLK}}$) and the required refresh rate of the SDRAM (f_{Refresh}). The following formula can be used:

$$RR = f_{\text{EMIF_CLK}} / f_{\text{Refresh}}$$

The SDRAM datasheet often communicates the required SDRAM Refresh Rate in terms of the number of REFR commands required in a given time interval. The required SDRAM Refresh Rate in the formula above can therefore be calculated by dividing the number of required cycles per time interval (n_{cycles}) by the time interval given in the datasheet ($t_{\text{Refresh Period}}$):

$$f_{\text{Refresh}} = n_{\text{cycles}} / t_{\text{Refresh Period}}$$

Combining these formulas, the value that should be programmed into the RR field can be computed as:

$$RR = f_{\text{EMIF_CLK}} \times t_{\text{Refresh Period}} / n_{\text{cycles}}$$

The following example illustrates calculating the value of RR. Given that:

- $f_{\text{EMIF_CLK}} = 100 \text{ MHz}$ (frequency of the EMIF clock)
- $t_{\text{Refresh Period}} = 64 \text{ ms}$ (required refresh interval of the SDRAM)
- $n_{\text{cycles}} = 8192$ (number of cycles in a refresh interval for the SDRAM)

RR can be calculated as:

$$RR = 100 \text{ MHz} \times 64 \text{ ms} / 8192$$

$$RR = 781.25$$

$$RR = 782 \text{ cycles} = 30\text{Eh cycles}$$

17.2.5.7 Self-Refresh Mode

The EMIF can be programmed to enter the self-refresh state by setting the SR bit of SDCR to 1. This will cause the EMIF to issue the SLFR command after completing any outstanding SDRAM access requests and clearing the refresh backlog counter by performing one or more auto refresh cycles. This places the attached SDRAM device into self-refresh mode in which it consumes a minimal amount of power while performing its own refresh cycles. The SR bit should be set and cleared using a byte-write to the upper byte of the SDRAM configuration register (SDCR) to avoid triggering the SDRAM initialization sequence.

While in the self-refresh state, the EMIF continues to service asynchronous bank requests and register accesses as normal, with one caveat. The EMIF will not park the data bus following a read to asynchronous memory while in the self-refresh state. Instead, the EMIF tri-states the data bus. Therefore, it is not recommended to perform asynchronous read operations while the EMIF is in the self-refresh state, in order to prevent floating inputs on the data bus. More information about data bus parking can be found in [Section 17.2.7](#).

The EMIF will exit from the self-refresh state if either of the following events occur:

- The SR bit of SDCR is cleared to 0.
- An SDRAM accesses is requested.

The EMIF exits from the self-refresh state by driving EMIF_CKE high and performing an auto refresh cycle.

The attached SDRAM device should also be placed into Self-Refresh Mode when changing the frequency of EMIF_CLK. If the frequency of EMIF_CLK changes while the SDRAM is not in Self-Refresh Mode, Procedure B in [Section 17.2.5.5](#) should be followed to reinitialize the device.

17.2.5.8 Power Down Mode

To support low-power modes, the EMIF can be requested to issue a POWER DOWN command to the SDRAM by setting the PD bit in the SDRAM configuration register (SDCR). When this bit is set, the EMIF will continue normal operation until all outstanding memory access requests have been serviced and the SDRAM refresh backlog (if there is one) has been cleared. At this point the EMIF will enter the power-down state. Upon entering this state, the EMIF will issue a POWER DOWN command (same as a NOP command but driving EMIF_CKE low on the same cycle). The EMIF then maintains EMIF_CKE low until it exits the power-down state.

Since the EMIF services the refresh backlog before it enters the power-down state, all internal banks of the SDRAM are closed (precharged) prior to issuing the POWER DOWN command. Therefore, the EMIF only supports Precharge Power Down. The EMIF does not support Active Power Down, where internal banks of the SDRAM are open (active) before the POWER DOWN command is issued.

During the power-down state, the EMIF services the SDRAM, asynchronous memory, and register accesses as normal, returning to the power-down state upon completion.

The PDWR bit in SDCR indicates whether the EMIF should perform refreshes in power-down state. If the PDWR bit is set, the EMIF exits the power-down state every time the Refresh Must level is set, performs AUTO REFRESH commands to the SDRAM, and returns back to the power-down state. This evenly distributes the refreshes to the SDRAM in power-down state. If the PDWR bit is not set, the EMIF does not perform any refreshes to the SDRAM. Therefore, the data integrity of the SDRAM is not assured upon power down exit if the PDWR bit is not set.

If the PD bit is cleared while in the power-down state, the EMIF will come out of the power-down state. The EMIF:

- Drives EMIF_CKE high.
- Enters its idle state.

17.2.5.9 SDRAM Read Operation

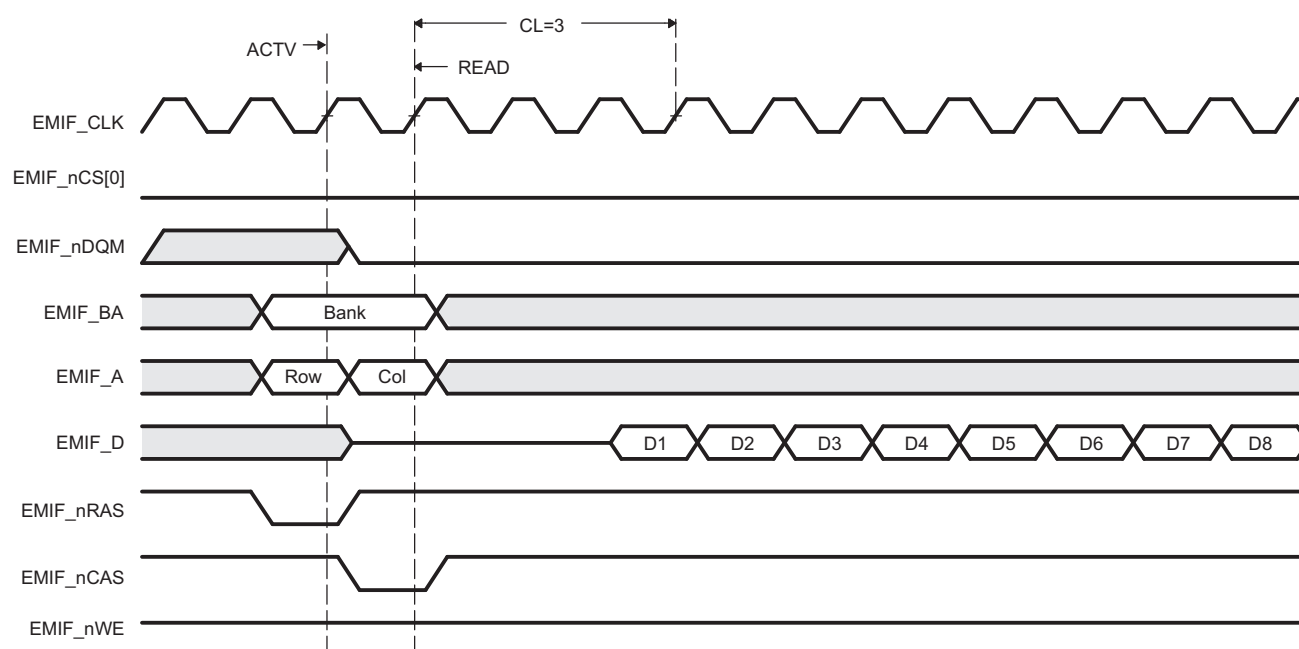
When the EMIF receives a read request to SDRAM from one of the requesters listed in [Section 17.2.2](#), it performs one or more read access cycles. A read access cycle begins with the issuing of the ACTV command to select the desired bank and row of the SDRAM device. After the row has been opened, the EMIF proceeds to issue a READ command while specifying the desired bank and column address. EMIF_A[10] is held low during the READ command to avoid auto-precharging. The READ command signals the SDRAM device to start bursting data from the specified address while the EMIF issues NOP commands. Following a READ command, the CL field of the SDRAM configuration register (SDCR) defines how many delay cycles will be present before the read data appears on the data bus. This is referred to as the CAS latency.

[Figure 17-5](#) shows the signal waveforms for a basic SDRAM read operation in which a burst of data is read from a single page. When the EMIF SDRAM interface is configured to 16 bit by setting the NM bit of the SDRAM configuration register (SDCR) to 1, a burst size of eight is used. [Figure 17-5](#) shows a burst size of eight.

The EMIF will truncate a series of bursting data if the remaining addresses of the burst are not required to complete the request. The EMIF can truncate the burst in three ways:

- By issuing another READ to the same page in the same bank.
- By issuing a PRE command in order to prepare for accessing a different page of the same bank.
- By issuing a BT command in order to prepare for accessing a page in a different bank.

Figure 17-5. Timing Waveform for Basic SDRAM Read Operation



Several other pins are also active during a read access. The EMIF_nDQM[1:0] pins are driven low during the READ commands and are kept low during the NOP commands that correspond to the burst request. The state of the other EMIF pins during each command can be found in [Table 17-5](#).

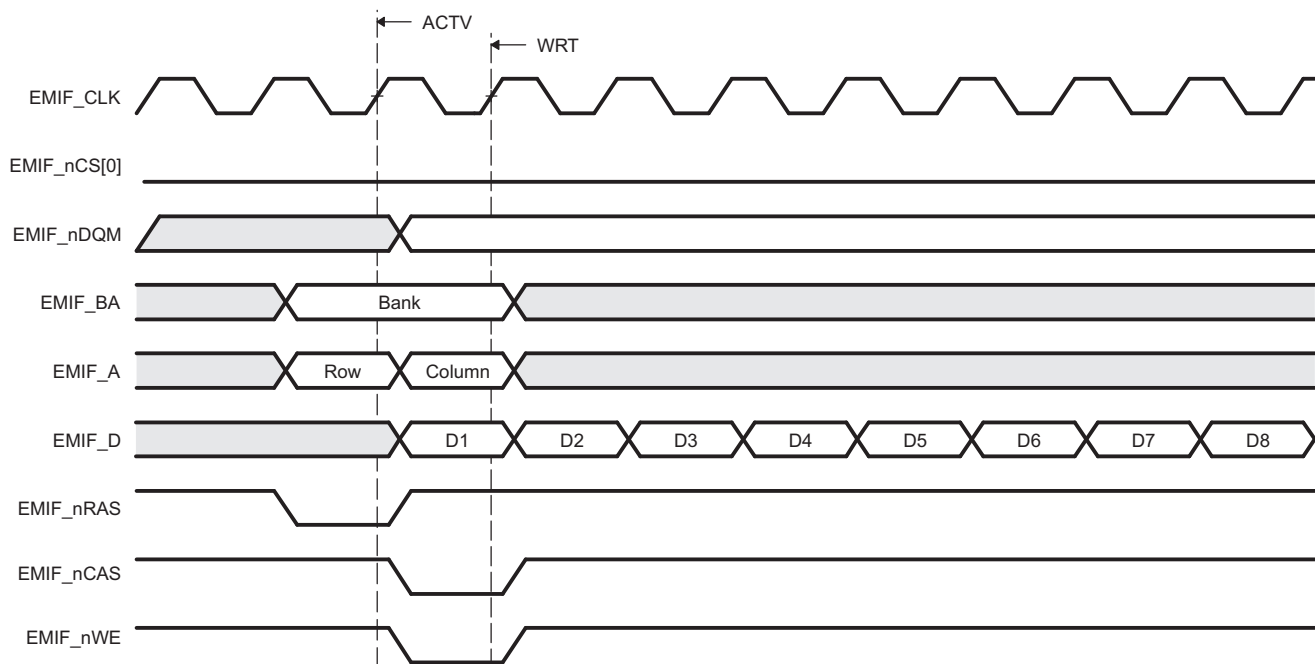
The EMIF schedules its commands based on the timing information that is provided to it in the SDRAM timing register (SDTIMR). The values for the timing parameters in this register should be chosen to satisfy the timing requirements listed in the SDRAM datasheet. The EMIF uses this timing information to avoid violating any timing constraints related to issuing commands. This is commonly accomplished by inserting NOP commands between various commands during an access. Refer to the register description of SDTIMR in [Section 17.3.6](#) for more details on the various timing parameters.

17.2.5.10 SDRAM Write Operations

When the EMIF receives a write request to SDRAM from one of the requesters listed in [Section 17.2.2](#), it performs one or more write-access cycles. A write-access cycle begins with the issuing of the ACTV command to select the desired bank and row of the SDRAM device. After the row has been opened, the EMIF proceeds to issue a WRT command while specifying the desired bank and column address. EMIF_A[10] is held low during the WRT command to avoid auto-precharging. The WRT command signals the SDRAM device to start writing a burst of data to the specified address while the EMIF issues NOP commands. The associated write data will be placed on the data bus in the cycle concurrent with the WRT command and with subsequent burst continuation NOP commands.

[Figure 17-6](#) shows the signal waveforms for a basic SDRAM write operation in which a burst of data is read from a single page. When the EMIF SDRAM interface is configured to 16-bit by setting the NM bit of the SDRAM configuration register (SDCR) to 1, a burst size of eight is used. [Figure 17-6](#) shows a burst size of eight.

Figure 17-6. Timing Waveform for Basic SDRAM Write Operation



The EMIF will truncate a series of bursting data if the remaining addresses of the burst are not part of the write request. The EMIF can truncate the burst in three ways:

- By issuing another WRT to the same page
- By issuing a PRE command in order to prepare for accessing a different page of the same bank
- By issuing a BT command in order to prepare for accessing a page in a different bank

Several other pins are also active during a write access. The EMIF_nDQM[1:0] pins are driven to select which bytes of the data word will be written to the SDRAM device. They are also used to mask out entire undesired data words during a burst access. The state of the other EMIF pins during each command can be found in [Table 17-5](#).

The EMIF schedules its commands based on the timing information that is provided to it in the SDRAM timing register (SDTIMR). The values for the timing parameters in this register should be chosen to satisfy the timing requirements listed in the SDRAM datasheet. The EMIF uses this timing information to avoid violating any timing constraints related to issuing commands. This is commonly accomplished by inserting NOP commands during various cycles of an access. Refer to the register description of SDTIMR in [Section 17.3.6](#) for more details on the various timing parameters.

17.2.5.11 Mapping from Logical Address to EMIF Pins

When the EMIF receives an SDRAM access request, it must convert the address of the access into the appropriate signals to send to the SDRAM device. The details of this address mapping are shown in [Table 17-13](#) for 16-bit operation. Using the settings of the IBANK and PAGESIZE fields of the SDRAM configuration register (SDCR), the EMIF determines which bits of the logical address will be mapped to the SDRAM row, column, and bank addresses.

As the logical address is incremented by one halfword (16-bit operation), the column address is likewise incremented by one until a page boundary is reached. When the logical address increments across a page boundary, the EMIF moves into the same page in the next bank of the attached device by incrementing the bank address EMIF_BA and resetting the column address. The page in the previous bank is left open until it is necessary to close it. This method of traversal through the SDRAM banks helps maximize the number of open banks inside of the SDRAM and results in an efficient use of the device. There is no limitation on the number of banks that can be open at one time, but only one page within a bank can be open at a time.

The EMIF uses the EMIF_nDQM[1:0] pins during a WRT command to mask out selected bytes or entire words. The EMIF_nDQM[1:0] pins are always low during a READ command.

Table 17-13. Mapping from Logical Address to EMIF Pins for 16-bit SDRAM

IBANK	PAGESIZE	Logical Address					
		12	11	10	9	8:1	0
0	0	Row Address				Column Address	EMIF_nDQM[0]
1	0	Row Address			EMIF_BA[0]	Column Address	EMIF_nDQM[0]
2	0	Row Address		EMIF_BA[1:0]		Column Address	EMIF_nDQM[0]
0	1	Row Address			Column Address		EMIF_nDQM[0]
1	1	Row Address		EMIF_BA[0]	Column Address		EMIF_nDQM[0]
2	1	Row Address	EMIF_BA[1:0]		Column Address		EMIF_nDQM[0]
0	2	Row Address		Column Address			EMIF_nDQM[0]
1	2	Row Address	EMIF_BA[0]	Column Address			EMIF_nDQM[0]
2	2	EMIF_BA[1:0]		Column Address			EMIF_nDQM[0]
0	3	Row Address	Column Address				EMIF_nDQM[0]
1	3	EMIF_BA[0]	Column Address				EMIF_nDQM[0]
2	3	EMIF_BA[0]	Column Address				EMIF_nDQM[0]

NOTE: The upper bit of the Row Address is used only when addressing 256-Mbit and 512-Mbit SDRAM memories.

17.2.6 Asynchronous Controller and Interface

The EMIF easily interfaces to a variety of asynchronous devices including NOR Flash and SRAM. It can be operated in two major modes (see [Table 17-14](#)):

- Normal Mode
- Select Strobe Mode

Table 17-14. Normal Mode vs. Select Strobe Mode

Mode	Function of EMIF_nDQM pins	Operation of EMIF_nCS[4:2]
Normal Mode	Byte enables	Active during the entire asynchronous access cycle
Select Strobe Mode	Byte enables	Active only during the strobe period of an access cycle

The first mode of operation is Normal Mode, in which the EMIF_nDQM pins of the EMIF function as byte enables. In this mode, the EMIF_nCS[4:2] pins behaves as typical chip select signals, remaining active for the duration of the asynchronous access. See [Section 17.2.6.1](#) for an example interface with multiple 8-bit devices.

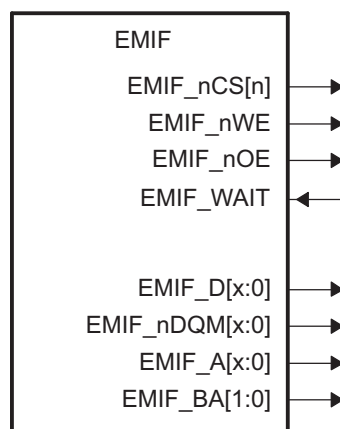
The second mode of operation is Select Strobe Mode, in which the EMIF_nCS[4:2] pins act as a strobe, active only during the strobe period of an access. In this mode, the EMIF_nDQM pins of the EMIF function as standard byte enables for reads and writes. A summary of the differences between the two modes of operation are shown in [Table 17-14](#). Refer to [Section 17.2.6.4](#) for the details of asynchronous operations in Normal Mode, and to [Section 17.2.6.5](#) for the details of asynchronous operations in Select Strobe Mode. The EMIF hardware defaults to Normal Mode, but can be manually switched to Select Strobe Mode by setting the SS bit in the asynchronous m ($m = 1, 2, 3$, or 4) configuration register (CE n CFG) ($n = 2, 3$, or 4). Throughout the chapter, m can hold the values 1, 2, 3 or 4; and n can hold the values 2, 3, or 4.

The EMIF also provides configurable cycle timing parameters and an Extended Wait Mode that allows the connected device to extend the strobe period of an access cycle. The following sections describe the features related to interfacing with external asynchronous devices.

17.2.6.1 Interfacing to Asynchronous Memory

[Figure 17-7](#) shows the EMIF's external pins used in interfacing with an asynchronous device. In EMIF_nCS[n], $n = 2, 3$, or 4 .

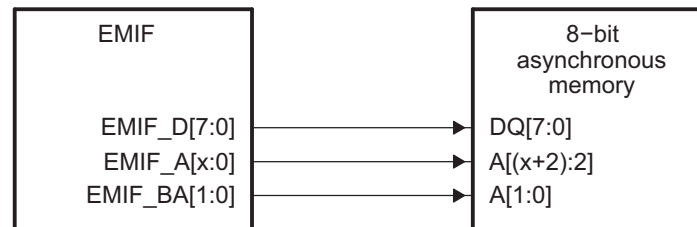
Figure 17-7. EMIF Asynchronous Interface



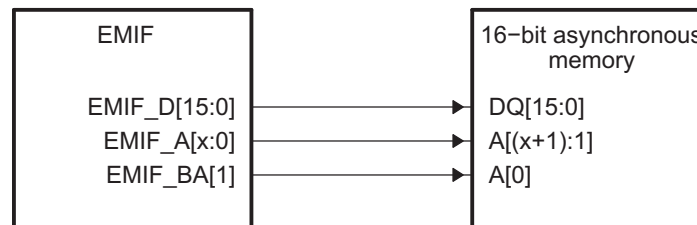
Of special note is the connection between the EMIF and the external device's address bus. The EMIF address pin EMIF_A[0] always provides the least-significant bit of a 32-bit word address. Therefore, when interfacing to a 16-bit or 8-bit asynchronous device, the EMIF_BA[1] and EMIF_BA[0] pins provide the least-significant bits of the halfword or byte address, respectively. Additionally, when the EMIF interfaces to a 16-bit asynchronous device, the EMIF_BA[0] pin can serve as the upper address line EMIF_A[12]. [Figure 17-8](#) and [Figure 17-9](#) show the mapping between the EMIF and the connected device's data and address pins for various programmed data bus widths. The data bus width may be configured in the asynchronous n configuration register (CENCFG).

[Figure 17-9](#) shows a common interface between the EMIF and external asynchronous memory. [Figure 17-9](#) shows an interface between the EMIF and an external memory with byte enables. The EMIF should be operated in either Normal Mode or Select Strobe Mode when using this interface, so that the EMIF_nDQM signals operate as byte enables.

Figure 17-8. EMIF to 8-bit/16-bit Memory Interface

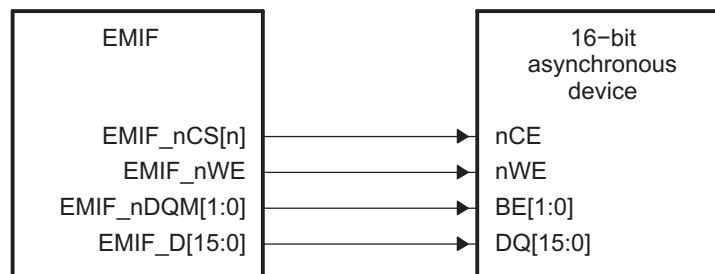


a) EMIF to 8-bit memory interface



b) EMIF to 16-bit memory interface

Figure 17-9. Common Asynchronous Interface



17.2.6.2 Accessing Larger Asynchronous Memories

The device has 13 dedicated EMIF address lines. If a device such as a large asynchronous flash needs to be attached to the EMIF, then GPIO pins may be used to control the flash device's upper address lines.

17.2.6.3 Configuring the EMIF for Asynchronous Accesses

The operation of the EMIF's asynchronous interface can be configured by programming the appropriate register fields. The reset value and bit position for each register field can be found in [Section 17.3](#). The following tables list the register fields that can be programmed and describe the purpose of each field. These registers can be programmed prior to accessing the external memory, and the transfer following a write to these registers will use the new configuration.

Table 17-15. Description of the Asynchronous *m* Configuration Register (CENCFG)

Parameter	Description
SS	Select Strobe mode. This bit selects the EMIF's mode of operation in the following way: - SS = 0 selects Normal Mode - EMIF_nDQM pins function as byte enables - EMIF_nCS[4:2] active for duration of access - SS = 1 selects Select Strobe Mode - EMIF_nDQM pins function as byte enables - EMIF_nCS[4:2] acts as a strobe.
EW	Extended Wait Mode enable. - EW = 0 disables Extended Wait Mode - EW = 1 enables Extended Wait Mode When set to 1, the EMIF enables its Extended Wait Mode in which the strobe width of an access cycle can be extended in response to the assertion of the EMIF_nWAIT pin. The WPn bit in the asynchronous wait cycle configuration register (AWCC) controls to polarity of EMIF_nWAIT pin. See Section 17.2.6.6 for more details on this mode of operation.
W_SETUP/R_SETUP	Read/Write setup widths. These fields define the number of EMIF clock cycles of setup time for the address pins (EMIF_A and EMIF_BA), byte enables (EMIF_nDQM), and asynchronous chip enable (EMIF_nCS[4:2]) before the read strobe pin (EMIF_nOE) or write strobe pin (EMIF_nWE) falls, minus one cycle. For writes, the W_SETUP field also defines the setup time for the data pins (EMIF_D). Refer to the datasheet of the external asynchronous device to determine the appropriate setting for this field.
W_STROBE/R_STROBE	Read/Write strobe widths. These fields define the number of EMIF clock cycles between the falling and rising of the read strobe pin (EMIF_nOE) or write strobe pin (EMIF_nWE), minus one cycle. If Extended Wait Mode is enabled by setting the EW field in the asynchronous <i>n</i> configuration register (CENCFG), these fields must be set to a value greater than zero. Refer to the datasheet of the external asynchronous device to determine the appropriate setting for this field.
W_HOLD/R_HOLD	Read/Write hold widths. These fields define the number of EMIF clock cycles of hold time for the address pins (EMIF_A and EMIF_BA), byte enables (EMIF_nDQM), and asynchronous chip enable (EMIF_nCS[4:2]) after the read strobe pin (EMIF_nOE) or write strobe pin (EMIF_nWE) rises, minus one cycle. For writes, the W_HOLD field also defines the hold time for the data pins (EMIF_D). Refer to the datasheet of the external asynchronous device to determine the appropriate setting for this field.
TA	Minimum turnaround time. This field defines the minimum number of EMIF clock cycles between asynchronous reads and writes, minus one cycle. The purpose of this feature is to avoid contention on the bus. The value written to this field also determines the number of cycles that will be inserted between asynchronous accesses and SDRAM accesses. Refer to the datasheet of the external asynchronous device to determine the appropriate setting for this field.

Table 17-15. Description of the Asynchronous *m* Configuration Register (CE_nCFG) (continued)

Parameter	Description
ASIZE	Asynchronous Device Bus Width. This field determines the data bus width of the asynchronous interface in the following way: - ASIZE = 0 selects an 8-bit bus - ASIZE = 1 selects a 16-bit bus The configuration of ASIZE determines the function of the EMIF_A and EMIF_BA pins as described in Section 17.2.6.1. This field also determines the number of external accesses required to fulfill a request generated by one of the sources mentioned in Section 17.2.2. For example, a request for a 32-bit word would require four external access when ASIZE = 0. Refer to the datasheet of the external asynchronous device to determine the appropriate setting for this field.

Table 17-16. Description of the Asynchronous Wait Cycle Configuration Register (AWCC)

Parameter	Description
WP _n	EM_WAIT Polarity. - WP_n = 0 selects active-low polarity - WP_n = 1 selects active-high polarity When set to 1, the EMIF will wait if the EMIF_nWAIT pin is high. When cleared to 0, the EMIF will wait if the EMIF_nWAIT pin is low. The EMIF must have the Extended Wait Mode enabled for the EMIF_nWAIT pin to affect the width of the strobe period.
MAX_EXT_WAIT	Maximum Extended Wait Cycles. This field configures the number of EMIF clock cycles the EMIF will wait for the EMIF_nWAIT pin to be deactivated during the strobe period of an access cycle. The maximum number of EMIF clock cycles it will wait is determined by the following formula: Maximum Extended Wait Cycles = (MAX_EXT_WAIT + 1) × 16 If the EMIF_nWAIT pin is not deactivated within the time specified by this field, the EMIF resumes the access cycle, registering whatever data is on the bus and proceeding to the hold period of the access cycle. This situation is referred to as an Asynchronous Timeout. An Asynchronous Timeout generates an interrupt, if it has been enabled in the EMIF interrupt mask set register (INTMSKSET). Refer to Section 17.2.9.1 for more information about the EMIF interrupts.

Table 17-17. Description of the EMIF Interrupt Mask Set Register (INTMSKSET)

Parameter	Description
WR_MASK_SET	Wait Rise Mask Set. Writing a 1 enables an interrupt to be generated when a rising edge on EMIF_nWAIT occurs
AT_MASK_SET	Asynchronous Timeout Mask Set. Writing a 1 to this bit enables an interrupt to be generated when an Asynchronous Timeout occurs.

Table 17-18. Description of the EMIF Interrupt Mast Clear Register (INTMSKCLR)

Parameter	Description
WR_MASK_CLR	Wait Rise Mask Clear. Writing a 1 to this bit disables the interrupt, clearing the WR_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET).
AT_MASK_CLR	Asynchronous Timeout Mask Clear. Writing a 1 to this bit prevents an interrupt from being generated when an Asynchronous Timeout occurs.

17.2.6.4 Read and Write Operations in Normal Mode

Normal Mode is the asynchronous interface's default mode of operation. It is selected when the SS bit in the asynchronous *n* configuration register (CEnCFG) is cleared to 0. In this mode, the EMIF_nDQM pins operate as byte enables. [Section 17.2.6.4.1](#) and [Section 17.2.6.4.2](#) explain the details of read and write operations while in Normal Mode.

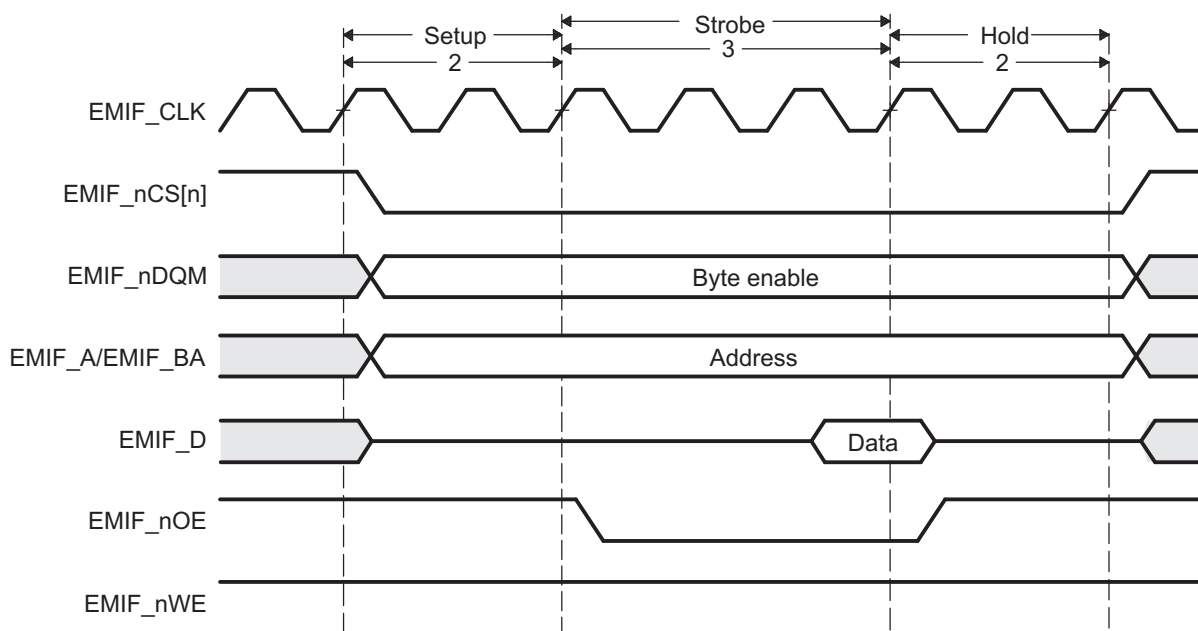
17.2.6.4.1 Asynchronous Read Operations (Normal Mode)

NOTE: During an entire asynchronous read operation, the EMIF_nWE pin is driven high.

An asynchronous read is performed when any of the requesters mentioned in [Section 17.2.2](#) request a read from the attached asynchronous memory. After the request is received, a read operation is initiated once it becomes the EMIF's highest priority task, according to the priority scheme detailed in [Section 17.2.13](#). In the event that the read request cannot be serviced by a single access cycle to the external device, multiple access cycles will be performed by the EMIF until the entire request is fulfilled. The details of an asynchronous read operation in Normal Mode are described in [Table 17-19](#). Also, [Figure 17-10](#) shows an example timing diagram of a basic read operation.

Table 17-19. Asynchronous Read Operation in Normal Mode

Time Interval	Pin Activity in Normal Mode
Turn-around period	Once the read operation becomes the highest priority task for the EMIF, the EMIF waits for the programmed number of turn-around cycles before proceeding to the setup period of the operation. The number of wait cycles is taken directly from the TA field of the asynchronous <i>n</i> configuration register (CEnCFG). There are two exceptions to this rule: - If the current read operation was directly preceded by another read operation, no turnaround cycles are inserted. - If the current read operation was directly preceded by a write operation and the TA field has been cleared to 0, one turn-around cycle will be inserted. After the EMIF has waited for the turnaround cycles to complete, it again checks to make sure that the read operation is still its highest priority task. If so, the EMIF proceeds to the setup period of the operation. If it is no longer the highest priority task, the EMIF terminates the operation.
Start of the setup period	The following actions occur at the start of the setup period: - The setup, strobe, and hold values are set according to the R_SETUP, R_STROBE, and R_HOLD values in CEnCFG. - The address pins EMIF_A and EMIF_BA become valid and carry the values described in Section 17.2.6.1. - EMIF_nCS[4:2] falls to enable the external device (if not already low from a previous operation)
Strobe period	The following actions occur during the strobe period of a read operation: - EMIF_nOE falls at the start of the strobe period - On the rising edge of the clock which is concurrent with the end of the strobe period: - EMIF_nOE rises - The data on the EMIF_D bus is sampled by the EMIF. In Figure 17-10, EMIF_nWAIT is inactive. If EMIF_nWAIT is instead activated, the strobe period can be extended by the external device to give it more time to provide the data. Section 17.2.6.6 contains more details on using the EMIF_nWAIT pin.
End of the hold period	At the end of the hold period: - The address pins EMIF_A and EMIF_BA become invalid - EMIF_nCS[4:2] rises (if no more operations are required to complete the current request) EMIF may be required to issue additional read operations to a device with a small data bus width in order to complete an entire word access. In this case, the EMIF immediately re-enters the setup period to begin another operation without incurring the turn-round cycle delay. The setup, strobe, and hold values are not updated in this case. If the entire word access has been completed, the EMIF returns to its previous state unless another asynchronous request has been submitted and is currently the highest priority task. If this is the case, the EMIF instead enters directly into the turnaround period for the pending read or write operation.

Figure 17-10. Timing Waveform of an Asynchronous Read Cycle in Normal Mode


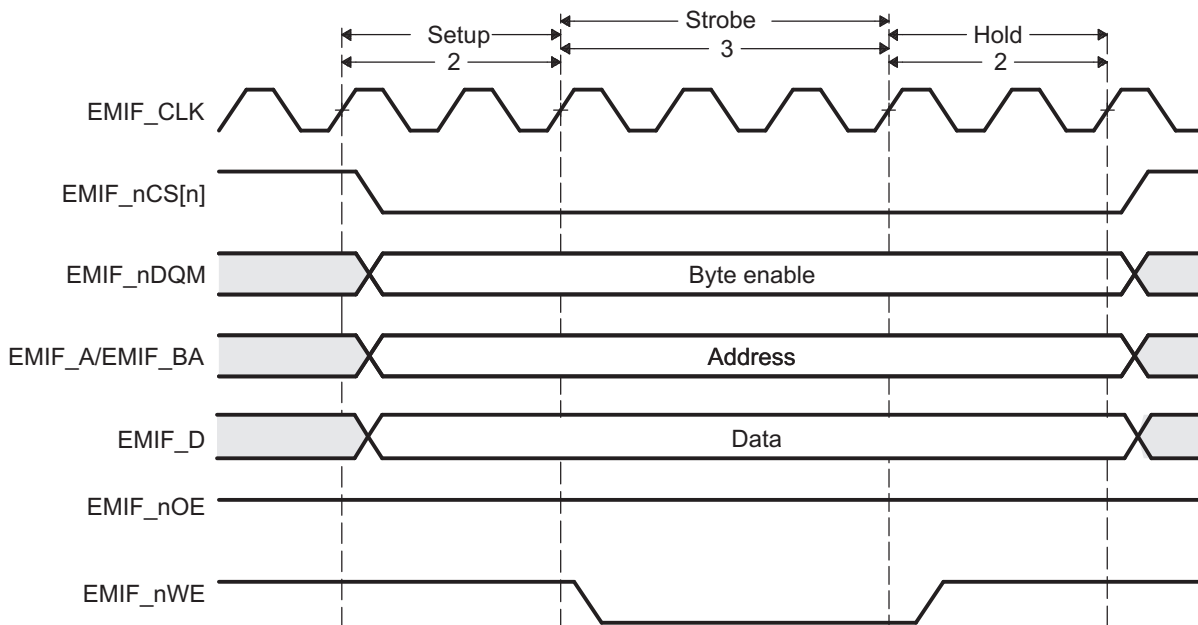
17.2.6.4.2 Asynchronous Write Operations (Normal Mode)

NOTE: During an entire asynchronous write operation, the EMIF_nOE pin is driven high.

An asynchronous write is performed when any of the requesters mentioned in [Section 17.2.2](#) request a write to memory in the asynchronous bank of the EMIF. After the request is received, a write operation is initiated once it becomes the EMIF's highest priority task, according to the priority scheme detailed in [Section 17.2.13](#). In the event that the write request cannot be serviced by a single access cycle to the external device, multiple access cycles will be performed by the EMIF until the entire request is fulfilled. The details of an asynchronous write operation in Normal Mode are described in [Table 17-20](#). Also, [Figure 17-11](#) shows an example timing diagram of a basic write operation.

Table 17-20. Asynchronous Write Operation in Normal Mode

Time Interval	Pin Activity in Normal Mode
Turnaround period	Once the write operation becomes the highest priority task for the EMIF, the EMIF waits for the programmed number of turn-around cycles before proceeding to the setup period of the operation. The number of wait cycles is taken directly from the TA field of the asynchronous <i>n</i> configuration register (CENCFG). There are two exceptions to this rule: • If the current write operation was directly preceded by another write operation, no turn-around cycles are inserted. • If the current write operation was directly preceded by a read operation and the TA field has been cleared to 0, one turnaround cycle will be inserted. After the EMIF has waited for the turn-around cycles to complete, it again checks to make sure that the write operation is still its highest priority task. If so, the EMIF proceeds to the setup period of the operation. If it is no longer the highest priority task, the EMIF terminates the operation.
Start of the setup period	The following actions occur at the start of the setup period: • The setup, strobe, and hold values are set according to the W_SETUP, W_STROBE, and W_HOLD values in CENCFG. • The address pins EMIF_A and EMIF_BA and the data pins EMIF_D become valid. The EMIF_A and EMIF_BA pins carry the values described in Section 17.2.6.1. • EMIF_nCS[4:2] falls to enable the external device (if not already low from a previous operation).
Strobe period	The following actions occur at the start of the strobe period of a write operation: 1. EMIF_nWE falls 2. The EMIF_nDQM pins become valid as byte enables. The following actions occur on the rising edge of the clock which is concurrent with the end of the strobe period: 1. EMIF_nWE rises 2. The EMIF_nDQM pins deactivate In Figure 17-11, EMIF_nWAIT is inactive. If EMIF_nWAIT is instead activated, the strobe period can be extended by the external device to give it more time to accept the data. Section 17.2.6.6 contains more details on using the EMIF_nWAIT pin.
End of the hold period	At the end of the hold period: • The address pins EMIF_A and EMIF_BA become invalid • The data pins become invalid • EMIF_nCS[n] (n = 2, 3, or 4) rises (if no more operations are required to complete the current request) The EMIF may be required to issue additional write operations to a device with a small data bus width in order to complete an entire word access. In this case, the EMIF immediately re-enters the setup period to begin another operation without incurring the turnaround cycle delay. The setup, strobe, and hold values are not updated in this case. If the entire word access has been completed, the EMIF returns to its previous state unless another asynchronous request has been submitted and is currently the highest priority task. If this is the case, the EMIF instead enters directly into the turnaround period for the pending read or write operation.

Figure 17-11. Timing Waveform of an Asynchronous Write Cycle in Normal Mode


17.2.6.5 Read and Write Operation in Select Strobe Mode

Select Strobe Mode is the EMIF's second mode of operation. It is selected when the SS bit of the asynchronous *n* configuration register (CEnCFG) is set to 1. In this mode, the EMIF_nDQM pins operate as byte enables and the EMIF_nCS[n] (*n* = 2, 3, or 4) pin is only active during the strobe period of an access cycle. [Section 17.2.6.4.1](#) and [Section 17.2.6.4.2](#) explain the details of read and write operations while in Select Strobe Mode.

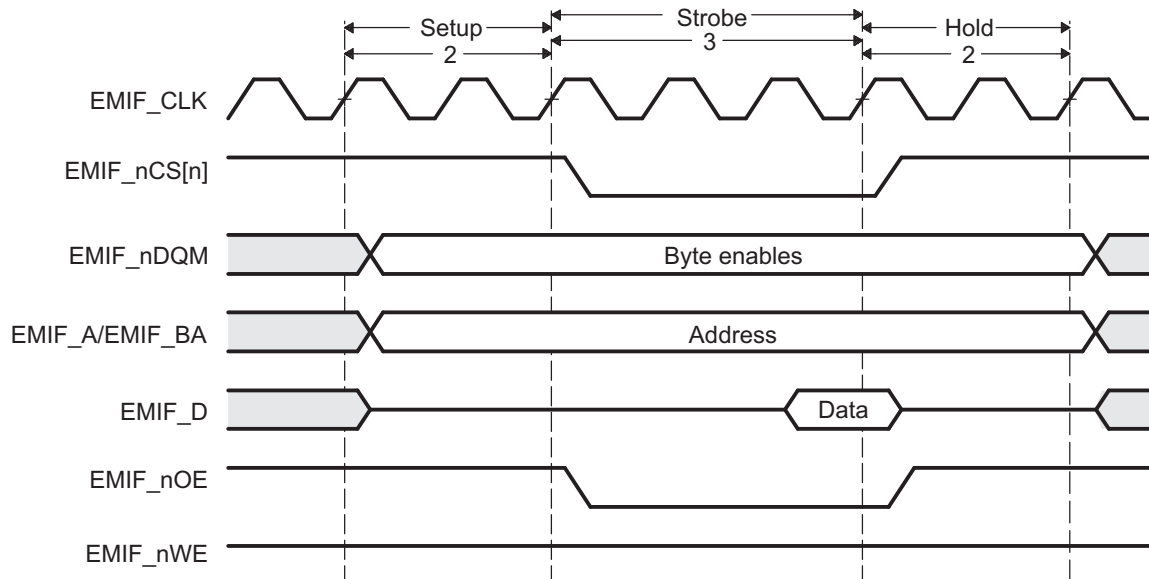
17.2.6.5.1 Asynchronous Read Operations (Select Strobe Mode)

NOTE: During the entirety of an asynchronous read operation, the EMIF_nWE pin is driven high.

An asynchronous read is performed when any of the requesters mentioned in [Section 17.2.2](#) request a read from the attached asynchronous memory. After the request is received, a read operation is initiated once it becomes the EMIF's highest priority task, according to the priority scheme detailed in [Section 17.2.13](#). In the event that the read request cannot be serviced by a single access cycle to the external device, multiple access cycles will be performed by the EMIF until the entire request is fulfilled. The details of an asynchronous read operation in Select Strobe Mode are described in [Table 17-21](#). Also, [Figure 17-12](#) shows an example timing diagram of a basic read operation.

Table 17-21. Asynchronous Read Operation in Select Strobe Mode

Time Interval	Pin Activity in Select Strobe Mode
Turnaround period	Once the read operation becomes the highest priority task for the EMIF, the EMIF waits for the programmed number of turn-around cycles before proceeding to the setup period of the operation. The number of wait cycles is taken directly from the TA field of the asynchronous <i>n</i> configuration register (CEnCFG). There are two exceptions to this rule: • If the current read operation was directly preceded by another read operation, no turn-around cycles are inserted. • If the current read operation was directly preceded by a write operation and the TA field has been cleared to 0, one turn-around cycle will be inserted. After the EMIF has waited for the turn-around cycles to complete, it again checks to make sure that the read operation is still its highest priority task. If so, the EMIF proceeds to the setup period of the operation. If it is no longer the highest priority task, the EMIF terminates the operation.
Start of the setup period	The following actions occur at the start of the setup period: • The setup, strobe, and hold values are set according to the R_SETUP, R_STROBE, and R_HOLD values in CEnCFG. • The address pins EMIF_A and EMIF_BA become valid and carry the values described in Section 17.2.6.1. • The EMIF_nDQM pins become valid as byte enables.
Strobe period	The following actions occur during the strobe period of a read operation: 1. EMIF_nCS[n] (<i>n</i> = 2, 3, or 4) and EMIF_nOE fall at the start of the strobe period 2. On the rising edge of the clock which is concurrent with the end of the strobe period: • EMIF_nCS[n] (<i>n</i> = 2, 3, or 4) and EMIF_nOE rise • The data on the EMIF_D bus is sampled by the EMIF. In Figure 17-12, EMIF_nWAIT is inactive. If EMIF_nWAIT is instead activated, the strobe period can be extended by the external device to give it more time to provide the data. Section 17.2.6.6 contains more details on using the EMIF_nWAIT pin.
End of the hold period	At the end of the hold period: • The address pins EMIF_A and EMIF_BA become invalid • The EMIF_nDQM pins become invalid The EMIF may be required to issue additional read operations to a device with a small data bus width in order to complete an entire word access. In this case, the EMIF immediately re-enters the setup period to begin another operation without incurring the turnaround cycle delay. The setup, strobe, and hold values are not updated in this case. If the entire word access has been completed, the EMIF returns to its previous state unless another asynchronous request has been submitted and is currently the highest priority task. If this is the case, the EMIF instead enters directly into the turnaround period for the pending read or write operation.

Figure 17-12. Timing Waveform of an Asynchronous Read Cycle in Select Strobe Mode


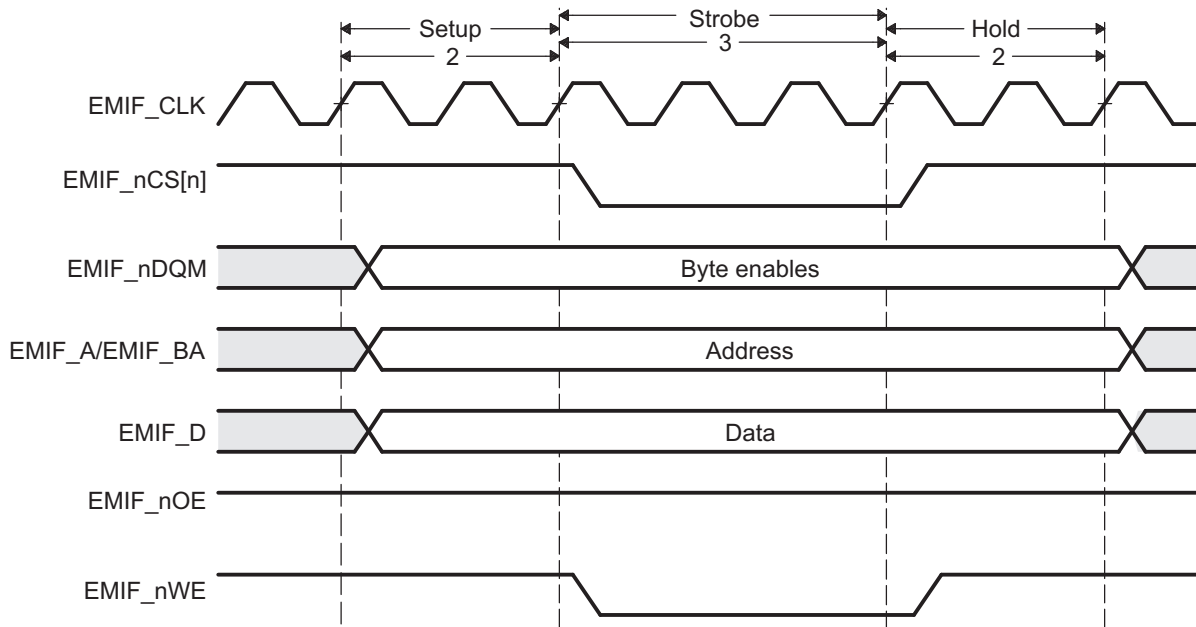
17.2.6.5.2 Asynchronous Write Operations (Select Strobe Mode)

NOTE: During the entirety of an asynchronous write operation, the EMIF_nOE pin is driven high.

An asynchronous write is performed when any of the requesters mentioned in [Section 17.2.2](#) request a write to memory in the asynchronous bank of the EMIF. After the request is received, a write operation is initiated once it becomes the EMIF's highest priority task, according to the priority scheme detailed in [Section 17.2.13](#). In the event that the write request cannot be serviced by a single access cycle to the external device, multiple access cycles will be performed by the EMIF until the entire request is fulfilled. The details of an asynchronous write operation in Select Strobe Mode are described in [Table 17-22](#). Also, [Figure 17-13](#) shows an example timing diagram of a basic write operation.

Table 17-22. Asynchronous Write Operation in Select Strobe Mode

Time Interval	Pin Activity in Select Strobe Mode
Turnaround period	Once the write operation becomes the highest priority task for the EMIF, the EMIF waits for the programmed number of turnaround cycles before proceeding to the setup period of the operation. The number of wait cycles is taken directly from the TA field of the asynchronous <i>n</i> configuration register (CEnCFG). There are two exceptions to this rule: - If the current write operation was directly preceded by another write operation, no turn-around cycles are inserted. - If the current write operation was directly preceded by a read operation and the TA field has been cleared to 0, one turnaround cycle will be inserted. After the EMIF has waited for the turnaround cycles to complete, it again checks to make sure that the write operation is still its highest priority task. If so, the EMIF proceeds to the setup period of the operation. If it is no longer the highest priority task, the EMIF terminates the operation.
Start of the setup period	The following actions occur at the start of the setup period: - The setup, strobe, and hold values are set according to the W_SETUP, W_STROBE, and W_HOLD values in CEnCFG. - The address pins EMIF_A and EMIF_BA and the data pins EMIF_D become valid. The EMIF_A and EMIF_BA pins carry the values described in Section 17.2.6.1. - The EMIF_nDQM pins become active as byte enables.
Strobe period	The following actions occur at the start of the strobe period of a write operation: - EMIF_nCS[n] (n = 2, 3, or 4) and EMIF_nWE fall The following actions occur on the rising edge of the clock which is concurrent with the end of the strobe period: - EMIF_nCS[n] (n = 2, 3, or 4) and EMIF_nWE rise In Figure 17-13, EMIF_nWAIT is inactive. If EMIF_nWAIT is instead activated, the strobe period can be extended by the external device to give it more time to accept the data. Section 17.2.6.6 contains more details on using the EMIF_nWAIT pin.
End of the hold period	At the end of the hold period: - The address pins EMIF_A and EMIF_BA become invalid - The data pins become invalid - The EMIF_nDQM pins become invalid The EMIF may be required to issue additional write operations to a device with a small data bus width in order to complete an entire word access. In this case, the EMIF immediately re-enters the setup period to begin another operation without incurring the turnaround cycle delay. The setup, strobe, and hold values are not updated in this case. If the entire word access has been completed, the EMIF returns to its previous state unless another asynchronous request has been submitted and is currently the highest priority task. If this is the case, the EMIF instead enters directly into the turn-around period for the pending read or write operation.

Figure 17-13. Timing Waveform of an Asynchronous Write Cycle in Select Strobe Mode


17.2.6.6 Extended Wait Mode and the EMIF_nWAIT Pin

The EMIF supports the Extend Wait Mode. This is a mode in which the external asynchronous device may assert control over the length of the strobe period. The Extended Wait Mode can be entered by setting the EW bit in the asynchronous n configuration register (CENCFG) ($n = 2, 3, \text{ or } 4$). When this bit is set, the EMIF monitors the EMIF_nWAIT pin to determine if the attached device wishes to extend the strobe period of the current access cycle beyond the programmed number of clock cycles.

When the EMIF detects that the EMIF_nWAIT pin has been asserted, it will begin inserting extra strobe cycles into the operation until the EMIF_nWAIT pin is deactivated by the external device. The EMIF will then return to the last cycle of the programmed strobe period and the operation will proceed as usual from this point. Please refer to the device data manual for details on the timing requirements of the EMIF_nWAIT signal.

The EMIF_nWAIT pin cannot be used to extend the strobe period indefinitely. The programmable MAX_EXT_WAIT field in the asynchronous wait cycle configuration register (AWCC) determines the maximum number of EMIF_CLK cycles the strobe period may be extended beyond the programmed length. When the counter expires, the EMIF proceeds to the hold period of the operation regardless of the state of the EMIF_nWAIT pin. The EMIF can also generate an interrupt upon expiration of this counter. See [Section 17.2.9.1](#) for details on enabling this interrupt.

For the EMIF to function properly in the Extended Wait mode, the WP n bit of AWCC must be programmed to match the polarity of the EMIF_nWAIT pin. In its reset state of 1, the EMIF will insert wait cycles when the EMIF_nWAIT pin is sampled high. When set to 0, the EMIF will insert wait cycles only when EMIF_nWAIT is sampled low. This programmability allows for a glueless connection to larger variety of asynchronous devices.

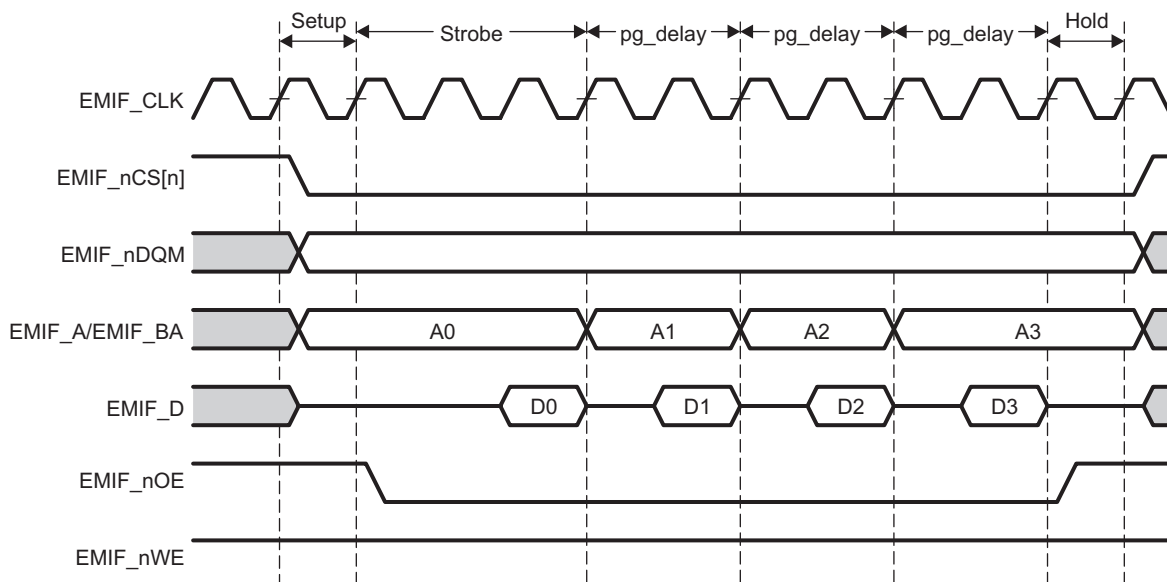
Finally, a restriction is placed on the strobe period timing parameters when operating in Extended Wait mode. Specifically, the sum of the W_SETUP and W_STROBE fields must be greater than 4, and the sum of the R_SETUP and R_STROBE fields must be greater than 4 for the EMIF to recognize the EMIF_nWAIT pin has been asserted. The W_SETUP, W_STROBE, R_SETUP, and R_STROBE fields are in CENCFG.

17.2.6.7 NOR Flash Page Mode

EMIF supports Page mode reads for NOR Flash on its asynchronous memory chip selects. This mode can be enabled by writing a 1 to the CS_n_PG_MD_EN ($n = 2, 3, \text{ or } 4$) field in the Page Mode Control register for the chip select in consideration. Whenever Page Mode for reads is enabled for a particular chip select, the page size for the device connected must also be programmed in the CS_n_PG_SIZE field of the Page Mode Control register. The address change to valid read data available timing must be programmed in the CS_n_PG_DEL field of the Page Control register. All other asynchronous memory timings must be programmed in the asynchronous configuration register (CENCFG). See [Figure 17-14](#) for read in asynchronous page mode.

NOTE: The Extended Wait mode and the Select Strobe mode must be disabled when using the asynchronous interface in Page mode.

Figure 17-14. Asynchronous Read in Page Mode



17.2.7 Data Bus Parking

The EMIF always drives the data bus to the previous write data value when it is idle. This feature is called data bus parking. Only when the EMIF issues a read command to the external memory does it stop driving the data bus. After the EMIF latches the last read data, it immediately parks the data bus again.

The one exception to this behavior occurs after performing an asynchronous read operation while the EMIF is in the self-refresh state. In this situation, the read operation is not followed by the EMIF parking the data bus. Instead, the EMIF tri-states the data bus. Therefore, it is not recommended to perform asynchronous read operations while the EMIF is in the self-refresh state, in order to prevent floating inputs on the data bus. External pull-ups, such as 10kΩ resistors, should be placed on the 16 EMIF data bus pins (which do not have internal pull-ups) if it is required to perform reads in this situation. The precise resistor value should be chosen so that the worst case combined off-state leakage currents do not cause the voltage levels on the associated pins to drop below the high-level input voltage requirement.

For information about the self-refresh state, see [Section 17.2.5.7](#).

17.2.8 Reset and Initialization Considerations

The EMIF memory controller has two active-low reset signals, `CHIP_RST_n` and `MOD_G_RST_n`. Both these reset signals are driven by the device system reset signal. This device does not offer the flexibility to reset just the EMIF state machine without also resetting the EMIF controller's memory-mapped registers. As soon as the device system reset is released (driven High), the EMIF memory controller immediately begins its initialization sequence. Command and data stored in the EMIF memory controller FIFOs are lost. Refer the Architecture chapter of the technical reference manual (TRM) for more information on conditions that can cause a device system reset to be asserted.

When system reset is released, the EMIF automatically begins running the SDRAM initialization sequence described in [Section 17.2.5.4](#). Even though the initialization procedure is automatic, a special procedure, found in [Section 17.2.5.5](#) must still be followed.

17.2.9 Interrupt Support

The EMIF supports a single interrupt to the CPU. [Section 17.2.9.1](#) details the generation and internal masking of EMIF interrupts.

17.2.9.1 Interrupt Events

There are three conditions that may cause the EMIF to generate an interrupt to the CPU. These conditions are:

- A rising edge on the `EMIF_nWAIT` signal (wait rise interrupt)
- An asynchronous time out
- Usage of unsupported addressing mode (line trap interrupt)

The wait rise interrupt occurs when a rising edge is detected on `EMIF_nWAIT` signal. This interrupt generation is not affected by the `WPn` bit in the asynchronous wait cycle configuration register (AWCC). The asynchronous time out interrupt condition occurs when the attached asynchronous device fails to deassert the `EMIF_nWAIT` pin within the number of cycles defined by the `MAX_EXT_WAIT` bit in AWCC (this happens only in extended wait mode). EMIF supports only linear incrementing and cache line wrap addressing modes. If an access request for an unsupported addressing mode is received, the EMIF will set the `LT` bit in the EMIF interrupt raw register (INTRAW) and treat the request as a linear incrementing request.

Only when the interrupt is enabled by setting the appropriate bit (`WR_MASK_SET/AT_MASK_SET/LT_MASK_SET`) in the EMIF interrupt mask set register (INTMSKSET) to 1, will the interrupt be sent to the CPU. Once enabled, the interrupt may be disabled by writing a 1 to the corresponding bit in the EMIF interrupt mask clear register (INTMSKCLR). The bit fields in both the INTMSKSET and INTMSKCLR may be used to indicate whether the interrupt is enabled. When the interrupt is enabled, the corresponding bit field in both the INTMSKSET and INTMSKCLR will have a value of 1; when the interrupt is disabled, the corresponding bit field will have a value of 0.

The EMIF interrupt raw register (INTRAW) and the EMIF interrupt mask register (INTMSK) indicate the status of each interrupt. The appropriate bit (`WR/AT/LT`) in INTRAW is set when the interrupt condition occurs, whether or not the interrupt has been enabled. However, the appropriate bit (`WR_MASKED/AT_MASKED/LT_MASKED`) in INTMSK is set only when the interrupt condition occurs and the interrupt is enabled. Writing a 1 to the bit in INTRAW clears the INTRAW bit as well as the corresponding bit in INTMSK. [Table 17-23](#) contains a brief summary of the interrupt status and control bit fields. See [Section 17.3](#) for complete details on the register fields.

Table 17-23. Interrupt Monitor and Control Bit Fields

Register Name	Bit Name	Description
EMIF interrupt raw register (INTRAW)	WR	This bit is set when an rising edge on the EMIF_nWAIT signal occurs. Writing a 1 clears the WR bit as well as the WR_MASKED bit in INTMSK.
	AT	This bit is set when an asynchronous timeout occurs. Writing a 1 clears the AT bit as well as the AT_MASKED bit in INTMSK.
	LT	This bit is set when an unsupported addressing mode is used. Writing a 1 clears LT bit as well as the LT_MASKED bit in INTMSK.
EMIF interrupt mask register (INTMSK)	WR_MASKED	This bit is set only when a rising edge on the EMIF_nWAIT signal occurs and the interrupt has been enabled by writing a 1 to the WR_MASK_SET bit in INTMSKSET.
	AT_MASKED	This bit is set only when an asynchronous timeout occurs and the interrupt has been enabled by writing a 1 to the AT_MASK_SET bit in INTMSKSET.
	LT_MASKED	This bit is set only when line trap interrupt occurs and the interrupt has been enabled by writing a 1 to the LT_MASK_SET bit in INTMSKSET.
EMIF interrupt mask set register (INTMSKSET)	WR_MASK_SET	Writing a 1 to this bit enables the wait rise interrupt.
	AT_MASK_SET	Writing a 1 to this bit enables the asynchronous timeout interrupt.
	LT_MASK_SET	Writing a 1 to this bit enables the line trap interrupt.
EMIF interrupt mask clear register (INTMSKCLR)	WR_MASK_CLR	Writing a 1 to this bit disables the wait rise interrupt.
	AT_MASK_CLR	Writing a 1 to this bit disables the asynchronous timeout interrupt.
	LT_MASK_CLR	Writing a 1 to this bit disables the line trap interrupt.

17.2.10 DMA Event Support

EMIF memory controller is a DMA slave peripheral and therefore does not generate DMA events. Data read and write requests may be made directly, by masters and the DMA.

17.2.11 EMIF Signal Multiplexing

For details on EMIF signal multiplexing, see the I/O Multiplexing Module chapter of the technical reference manual.

17.2.12 Memory Map

For information describing the device memory-map, see your device-specific datasheet.

17.2.13 Priority and Arbitration

[Section 17.2.2](#) describes the external prioritization and arbitration among requests from different sources within the microcontroller. The result of this external arbitration is that only one request is presented to the EMIF at a time. Once the EMIF completes a request, the external arbiter then provides the EMIF with the next pending request.

Internally, the EMIF undertakes memory device transactions according to a strict priority scheme. The highest priority events are:

- A device reset.
- A write to any of the three least significant bytes of the SDRAM configuration register (SDCR).

Either of these events will cause the EMIF to immediately commence its initialization sequence as described in [Section 17.2.5.4](#).

Once the EMIF has completed its initialization sequence, it performs memory transactions according to the following priority scheme (highest priority listed first):

1. If the EMIF's backlog refresh counter is at the Refresh Must urgency level, the EMIF performs multiple SDRAM auto refresh cycles until the Refresh Release urgency level is reached.
2. If an SDRAM or asynchronous read has been requested, the EMIF performs a read operation.
3. If the EMIF's backlog refresh counter is at the Refresh Need urgency level, the EMIF performs an SDRAM auto refresh cycle.
4. If an SDRAM or asynchronous write has been requested, the EMIF performs a write operation.
5. If the EMIF's backlog refresh counter is at the Refresh May or Refresh Release urgency level, the EMIF performs an SDRAM auto refresh cycle.
6. If the value of the SR bit in SDCR has been set to 1, the EMIF will enter the self-refresh state as described in [Section 17.2.5.7](#).

After taking one of the actions listed above, the EMIF then returns to the top of the priority list to determine its next action.

Because the EMIF does not issue auto-refresh cycles when in the self-refresh state, the above priority scheme does not apply when in this state. See [Section 17.2.5.7](#) for details on the operation of the EMIF when in the self-refresh state.

17.2.14 System Considerations

This section describes various system considerations to keep in mind when operating the EMIF.

17.2.14.1 Asynchronous Request Times

In a system that interfaces to both SDRAM and asynchronous memory, the asynchronous requests must not take longer than the smaller of the following two values:

- t_{RAS} (typically 120 μ s) - to avoid violating the maximum time allowed between issuing an ACTV and PRE command to the SDRAM.
- $t_{Refresh\ Rate} \times 11$ (typically 15.7 μ s $\times$ 11 = 172.7 μ s) - to avoid refresh violations on the SDRAM.

The length of an asynchronous request is controlled by multiple factors, the primary factor being the number of access cycles required to complete the request. For example, an asynchronous request for 4 bytes will require four access cycles using an 8-bit data bus and only two access cycle using a 16-bit data bus. The maximum request size that the EMIF can be sent is 16 words, therefore the maximum number of access cycles per memory request is 64 when the EMIF is configured with an 8-bit data bus. The length of the individual access cycles that make up the asynchronous request is determined by the programmed setup, strobe, hold, and turnaround values, but can also be extended with the assertion of the EMIF_nWAIT input signal up to a programmed maximum limit. It is up to the user to make sure that an entire asynchronous request does not exceed the timing values listed above when also interfacing to an SDRAM device. This can be done by limiting the asynchronous timing parameters.

17.2.14.2 Interface to External Peripheral or FIFO Memory

If EMIF is used to interface to an external peripheral or FIFO logic (for example, UHPI), it is recommended to use the host CPU's Memory Protection Unit (MPU) to define this external memory range as a region that is either strongly-ordered or of device type.

17.2.14.3 Interface to External SDRAM

If EMIF is used to interface to an external SDRAM, it is recommended to burst as much as possible to normal memory to improve the interface bandwidth.

17.2.15 Power Management

Power dissipation from the EMIF memory controller may be managed by following methods:

- Self-refresh mode
- Power-down mode
- Gating input clocks to the module off

Gating input clocks off to the EMIF memory controller achieves higher power savings when compared to the power savings of self-refresh or power down mode. The input clock VCLK3 can be turned off through the use of the Global Clock Module (GCM). Before gating clocks off, the EMIF memory controller must place the SDR SDRAM memory in self-refresh mode. If the external memory requires a continuous clock, the VCLK3 clock domain must not be turned off because this may result in data corruption. See the following subsections for the proper procedures to follow when stopping the EMIF memory controller clocks.

17.2.15.1 Power Management Using Self-Refresh Mode

The EMIF can be placed into a self-refresh state in order to place the attached SDRAM devices into self-refresh mode, which consumes less power for most SDRAM devices. In this state, the attached SDRAM device uses an internal clock to perform its own auto refresh cycles. This maintains the validity of the data in the SDRAM without the need for any external commands. Refer to [Section 17.2.5.7](#) for more details on placing the EMIF into the self-refresh state.

17.2.15.2 Power Management Using Power Down Mode

In the power down mode, EMIF drives EMIF_CKE low to lower the power consumption. EMIF_CKE goes high when there is a need to send refresh (REFR) commands, after which EMIF_CKE is again driven low. EMIF_CKE remains low until any request arrives. Refer to [Section 17.2.5.8](#) for more details on placing EMIF in power down mode.

17.2.16 Emulation Considerations

EMIF memory controller remains fully functional during emulation halts in order to allow emulation access to external memory.

17.3 EMIF Registers

The external memory interface (EMIF) is controlled by programming its internal memory-mapped registers (MMRs). [Table 17-24](#) lists the memory-mapped registers for the EMIF.

NOTE: All EMIF MMRs, except SDCR, support only word (32-bit) accesses. Performing a byte (8-bit) or halfword (16-bit) write to these registers results in undefined behavior. The SDCR is byte writable to allow the setting of the SR, PD, and PDWR bits without triggering the SDRAM initialization sequence.

The EMIF registers must always be accessed using 32-bit accesses (unless otherwise specified in this chapter). The base address of the EMIF memory-mapped registers is FCFF E800h.

Table 17-24. External Memory Interface (EMIF) Registers

Offset	Acronym	Register Description	Section
00h	MIDR	Module ID Register	Section 17.3.1
04h	AWCC	Asynchronous Wait Cycle Configuration Register	Section 17.3.2
08h	SDCR	SDRAM Configuration Register	Section 17.3.3
0Ch	SDRCR	SDRAM Refresh Control Register	Section 17.3.4
10h	CE2CFG	Asynchronous 1 Configuration Register	Section 17.3.5
14h	CE3CFG	Asynchronous 2 Configuration Register	Section 17.3.5
18h	CE4CFG	Asynchronous 3 Configuration Register	Section 17.3.5
1Ch	CE5CFG	Asynchronous 4 Configuration Register	Section 17.3.5
20h	SDTIMR	SDRAM Timing Register	Section 17.3.6
3Ch	SDSRETR	SDRAM Self Refresh Exit Timing Register	Section 17.3.7
40h	INTRAW	EMIF Interrupt Raw Register	Section 17.3.8
44h	INTMSK	EMIF Interrupt Mask Register	Section 17.3.9
48h	INTMSKSET	EMIF Interrupt Mask Set Register	Section 17.3.10
4Ch	INTMSKCLR	EMIF Interrupt Mask Clear Register	Section 17.3.11
68h	PMCR	Page Mode Control Register	Section 17.3.12

17.3.1 Module ID Register (MIDR)

This is a read-only register indicating the module ID of the EMIF. The MIDR is shown in [Figure 17-15](#) and described in [Table 17-25](#).

Figure 17-15. Module ID Register (MIDR) [offset = 00]

31	0
REV	
R-x	

LEGEND: R = Read only; -n = value after reset

Table 17-25. Module ID Register (MIDR) Field Descriptions

Bit	Field	Value	Description
31-0	REV	x	Module ID of EMIF. See the device-specific data manual.

17.3.2 Asynchronous Wait Cycle Configuration Register (AWCC)

The asynchronous wait cycle configuration register (AWCC) is used to configure the parameters for extended wait cycles. Both the polarity of the EMIF_nWAIT pin(s) and the maximum allowable number of extended wait cycles can be configured. The AWCC is shown in Figure 17-16 and described in Table 17-26. Not all devices support both EMIF_nWAIT[1] and EMIF_nWAIT[0], see the device-specific data manual to determine support on each device.

NOTE: The EW bit in the asynchronous *n* configuration register (CEnCFG) must be set to allow for the insertion of extended wait cycles.

Figure 17-16. Asynchronous Wait Cycle Configuration Register (AWCCR) [offset = 04h]

31	30	29	28	27	24	23	22	21	20	19	18	17	16
Reserved	WP1	WP0	Reserved	Reserved	CS5_WAIT	CS4_WAIT	CS3_WAIT	CS2_WAIT					
R-3h	R/W-1	R/W-1	R-0	R-0	R/W-0	R/W-0	R/W-0	R/W-0					
15				8	7								0
Reserved					MAX_EXT_WAIT								
R-0					R/W-80h								

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Table 17-26. Asynchronous Wait Cycle Configuration Register (AWCCR) Field Descriptions

Bit	Field	Value	Description
31-30	Reserved	3h	Reserved
29	WP1	0 1	EMIF_nWAIT[1] polarity bit. This bit defines the polarity of the EMIF_nWAIT[1] pin. Insert wait cycles if EMIF_nWAIT[1] pin is low. Insert wait cycles if EMIF_nWAIT[1] pin is high.
28	WP0	0 1	EMIF_nWAIT[0] polarity bit. This bit defines the polarity of the EMIF_nWAIT[0] pin. Insert wait cycles if EMIF_nWAIT[0] pin is low. Insert wait cycles if EMIF_nWAIT[0] pin is high.
27-24	Reserved	0	Reserved
23-22	CS5_WAIT	0-3h	Chip Select 5 WAIT signal selection. This signal determines which EMIF_nWAIT[<i>n</i>] signal used for memory accesses to chip select 5 memory space. This device does not support chip select 5, so any value written to this field has no effect.
21-20	CS4_WAIT	0 1h 2h-3h	Chip Select 4 WAIT signal selection. This signal determines which EMIF_nWAIT[<i>n</i>] signal used for memory accesses to chip select 4 memory space. EMIF_nWAIT[0] pin is used to control external wait states. EMIF_nWAIT[1] pin is used to control external wait states. Reserved
19-18	CS3_WAIT	0 1h 2h-3h	Chip Select 3 WAIT signal selection. This signal determines which EMIF_nWAIT[<i>n</i>] signal used for memory accesses to chip select 3 memory space. EMIF_nWAIT[0] pin is used to control external wait states. EMIF_nWAIT[1] pin is used to control external wait states. Reserved
17-16	CS2_WAIT	0 1h 2h-3h	Chip Select 2 WAIT signal selection. This signal determines which EMIF_nWAIT[<i>n</i>] signal used for memory accesses to chip select 2 memory space. EMIF_nWAIT[0] pin is used to control external wait states. EMIF_nWAIT[1] pin is used to control external wait states. Reserved
15-8	Reserved	0	Reserved
7-0	MAX_EXT_WAIT	0-FFh	Maximum extended wait cycles. The EMIF waits for a maximum of (MAX_EXT_WAIT + 1) × 16 clock cycles before it stops inserting asynchronous wait cycles and proceeds to the hold period of the access.

17.3.3 SDRAM Configuration Register (SDCR)

The SDRAM configuration register (SDCR) is used to configure various parameters of the SDRAM controller such as the number of internal banks, the internal page size, and the CAS latency to match those of the attached SDRAM device. In addition, this register is used to put the attached SDRAM device into Self-Refresh mode. The SDCR is shown in [Figure 17-17](#) and described in [Table 17-27](#).

NOTE: Writing to the lower three bytes of this register causes the EMIF to start the SDRAM initialization sequence described in [Section 17.2.5.4](#).

Figure 17-17. SDRAM Configuration Register (SDCR) [offset = 08h]

31	30	29	28	24
SR	PD	PDWR	Reserved	
R/W-0	R/W-0	R/W-0	R-0	
23	16			
Reserved				
R-0				
15	14	13	12	11
Reserved	NM ^(A)	Reserved	CL	BIT11_9LOCK
R-0	R/W-0	R-0	R/W-3h	R/W-0
7	6	4	3	2
Reserved	IBANK		Reserved	PAGESIZE
R-0	R/W-2h		R-0	R/W-0

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

A. The NM bit must be set to 1, if the EMIF on your device only has 16 data bus pins.

Table 17-27. SDRAM Configuration Register (SDCR) Field Descriptions

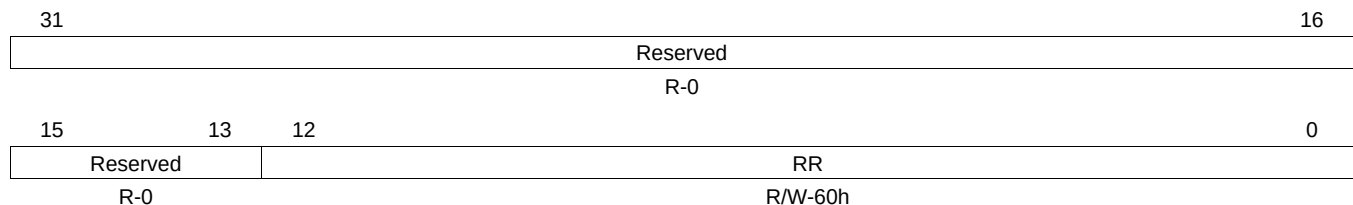
Bit	Field	Value	Description
31	SR	0 1	Self-Refresh mode bit. This bit controls entering and exiting of the Self-Refresh mode described in Section 17.2.5.7 . The field should be written using a byte-write to the upper byte of SDCR to avoid triggering the SDRAM initialization sequence. Writing a 0 to this bit causes connected SDRAM devices and the EMIF to exit the Self-Refresh mode. Writing a 1 to this bit causes connected SDRAM devices and the EMIF to enter the Self-Refresh mode.
30	PD	0 1	Power Down bit. This bit controls entering and exiting of the power-down mode. The field should be written using a byte-write to the upper byte of SDCR to avoid triggering the SDRAM initialization sequence. If both SR and PD bits are set, the EMIF goes into Self Refresh. Writing a 0 to this bit causes connected SDRAM devices and the EMIF to exit the power-down mode. Writing a 1 to this bit causes connected SDRAM devices and the EMIF to enter the power-down mode.
29	PDWR		Perform refreshes during power down. Writing a 1 to this bit causes EMIF to exit power-down state and issue and AUTO REFRESH command every time Refresh May level is set.
28-15	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
14	NM	0 1	Narrow mode bit. This bit defines whether a 16- or 32-bit-wide SDRAM is connected to the EMIF. This bit field must always be set to 1. Writing to this field triggers the SDRAM initialization sequence. 32-bit SDRAM data bus is used. 16-bit SDRAM data bus is used.
13-12	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.

Table 17-27. SDRAM Configuration Register (SDCR) Field Descriptions (continued)

Bit	Field	Value	Description
11-9	CL	0-1h Reserved 2h CAS latency = 2 EMIF_CLK cycles 3h CAS latency = 3 EMIF_CLK cycles 4h-7h Reserved	CAS Latency. This field defines the CAS latency to be used when accessing connected SDRAM devices. A 1 must be simultaneously written to the BIT11_9LOCK bit field of this register in order to write to the CL bit field. Writing to this field triggers the SDRAM initialization sequence.
8	BIT11_9LOCK	0 CL cannot be written. 1 CL can be written.	Bits 11 to 9 lock. CL can only be written if BIT11_9LOCK is simultaneously written with a 1. BIT11_9LOCK is always read as 0. Writing to this field triggers the SDRAM initialization sequence.
7	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
6-4	IBANK	0 1 bank SDRAM devices. 1 2 bank SDRAM devices. 2 4 bank SDRAM devices. 3h-7h Reserved.	Internal SDRAM Bank size. This field defines number of banks inside the connected SDRAM devices. Writing to this field triggers the SDRAM initialization sequence.
3	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
2-0	PAGESIZE	0 8 column address bits (256 elements per row). 1h 9 column address bits (512 elements per row). 2h 10 column address bits (1024 elements per row). 3h 11 column address bits (2048 elements per row). 4h-7h Reserved	Page Size. This field defines the internal page size of connected SDRAM devices. Writing to this field triggers the SDRAM initialization sequence.

17.3.4 SDRAM Refresh Control Register (SDRCR)

The SDRAM refresh control register (SDRCR) is used to configure the rate at which connected SDRAM devices are automatically refreshed by the EMIF. Refer to [Section 17.2.5.6](#) on the refresh controller for more details. The SDRCR is shown in [Figure 17-18](#) and described in [Table 17-28](#).

Figure 17-18. SDRAM Refresh Control Register (SDRCR) [offset = 0Ch]


LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 17-28. SDRAM Refresh Control Register (SDRCR) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
12-0	RR	0-1FFFh	Refresh Rate. This field is used to define the SDRAM refresh period in terms of EMIF_CLK cycles. Writing a value < 0x0020 to this field causes it to be loaded with (2 × T_RFC) + 1 value from the SDRAM timing register (SDTIMR).

17.3.5 Asynchronous *n* Configuration Registers (CE2CFG-CE5CFG)

The asynchronous *n* configuration registers (CE2CFG, CE3CFG, CE4CFG, and CE5CFG) are used to configure the shaping of the address and control signals during an access to asynchronous memory connected to CS2, CS3, CS4, and CS5, respectively. CS5 is not available on this device. It is also used to program the width of asynchronous interface and to select from various modes of operation. This register can be written prior to any transfer, and any asynchronous transfer following the write uses the new configuration. The CE n CFG is shown in [Figure 17-19](#) and described in [Table 17-29](#).

Figure 17-19. Asynchronous *n* Configuration Register (CE n CFG) [offset = 10h - 1Ch]

31	30	29	26	25	24
SS	EW ^(A)	W_SETUP		W_STROBE ^(B)	
R/W-0	R/W-0	R/W-Fh		R/W-3Fh	
23	20	19	17	16	
W_STROBE ^(B)		W_HOLD		R_SETUP	
R/W-3Fh		R/W-7h		R/W-Fh	
15	13	12	7	6	4
R_SETUP	R_STROBE ^(B)		R_HOLD	TA	ASIZE
R/W-Fh	R/W-3Fh		R/W-7h	R/W-3h	R/W-0

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

A. The EW bit must be cleared to 0.

B. This bit field must be cleared to 0, if the EMIF on your device does not have an EMIF_nWAIT pin.

Table 17-29. Asynchronous *n* Configuration Register (CE n CFG) Field Descriptions

Bit	Field	Value	Description
31	SS	0 1	Select Strobe bit. This bit defines whether the asynchronous interface operates in Normal Mode or Select Strobe Mode. See Section 17.2.6 for details on the two modes of operation. Normal Mode is enabled. Select Strobe Mode is enabled.
30	EW	0 1	Extend Wait bit. This bit defines whether extended wait cycles are enabled. See Section 17.2.6.6 on extended wait cycles for details. This bit field must be cleared to 0, if the EMIF on your device does not have an EMIF_nWAIT pin. Extended wait cycles are disabled. Extended wait cycles are enabled.
29-26	W_SETUP	0-Fh	Write setup width in EMIF_CLK cycles, minus one cycle. See Section 17.2.6.3 for details.
25-20	W_STROBE	0-3Fh	Write strobe width in EMIF_CLK cycles, minus one cycle. See Section 17.2.6.3 for details.
19-17	W_HOLD	0-7h	Write hold width in EMIF_CLK cycles, minus one cycle. See Section 17.2.6.3 for details.
16-13	R_SETUP	0-Fh	Read setup width in EMIF_CLK cycles, minus one cycle. See Section 17.2.6.3 for details.
12-7	R_STROBE	0-3Fh	Read strobe width in EMIF_CLK cycles, minus one cycle. See Section 17.2.6.3 for details.
6-4	R_HOLD	0-7h	Read hold width in EMIF_CLK cycles, minus one cycle. See Section 17.2.6.3 for details.
3-2	TA	0-3h	Minimum Turn-Around time. This field defines the minimum number of EMIF_CLK cycles between reads and writes, minus 1 cycle. See Section 17.2.6.3 for details.
1-0	ASIZE	0 1h 2h-3h	Asynchronous Data Bus Width. This field defines the width of the asynchronous device data bus. 8-bit data bus 16-bit data bus Reserved

17.3.6 SDRAM Timing Register (SDTIMR)

The SDRAM timing register (SDTIMR) is used to program many of the SDRAM timing parameters. Consult the SDRAM datasheet for information on the appropriate values to program into each field. The SDTIMR is shown in [Figure 17-20](#) and described in [Table 17-30](#).

Figure 17-20. SDRAM Timing Register (SDTIMR) [offset = 20h]

31	27	26	24	23	22	20	19	18	16
T_RFC		T_RP		Rsvd	T_RCD		Rsvd	T_WR	
R/W-8h		R/W-2h		R-0	R/W-2h		R-0	R/W-1h	
15	12	11	8	7	6	4	3	0	
T_RAS		T_RC		Rsvd	T_RRD		Reserved		
R/W-5h		R/W-8h		R-0	R/W-1h		R-0		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

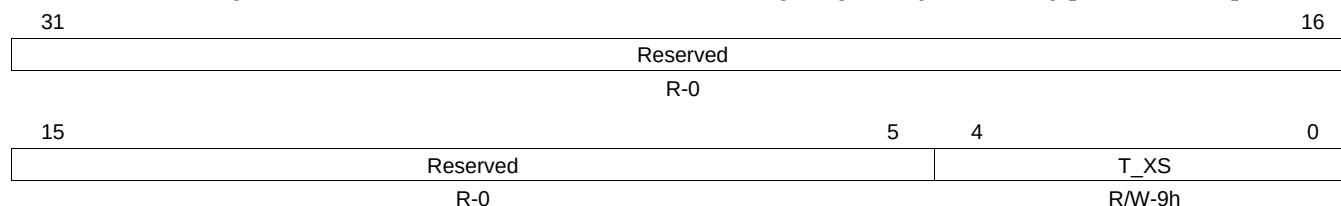
Table 17-30. SDRAM Timing Register (SDTIMR) Field Descriptions

Bit	Field	Value	Description
31-27	T_RFC	0-1Fh	Specifies the Trfc value of the SDRAM. This defines the minimum number of EMIF_CLK cycles from Refresh (REFR) to Refresh (REFR), minus 1: $T_RFC = (Trfc/t_{EMIF_CLK}) - 1$
26-24	T_RP	0-7h	Specifies the Trp value of the SDRAM. This defines the minimum number of EMIF_CLK cycles from Precharge (PRE) to Activate (ACTV) or Refresh (REFR) command, minus 1: $T_RP = (Trp/t_{EMIF_CLK}) - 1$
23	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
22-20	T_RCD	0-7h	Specifies the Trcd value of the SDRAM. This defines the minimum number of EMIF_CLK cycles from Active (ACTV) to Read (READ) or Write (WRT), minus 1: $T_RCD = (Trcd/t_{EMIF_CLK}) - 1$
19	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
18-16	T_WR	0-7h	Specifies the Twr value of the SDRAM. This defines the minimum number of EMIF_CLK cycles from last Write (WRT) to Precharge (PRE), minus 1: $T_WR = (Twr/t_{EMIF_CLK}) - 1$
15-12	T_RAS	0-Fh	Specifies the Tras value of the SDRAM. This defines the minimum number of EMIF_CLK clock cycles from Activate (ACTV) to Precharge (PRE), minus 1: $T_RAS = (Tras/t_{EMIF_CLK}) - 1$
11-8	T_RC	0-Fh	Specifies the Trc value of the SDRAM. This defines the minimum number of EMIF_CLK clock cycles from Activate (ACTV) to Activate (ACTV), minus 1: $T_RC = (Trc/t_{EMIF_CLK}) - 1$
7	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
6-4	T_RRD	0-7h	Specifies the Trrd value of the SDRAM. This defines the minimum number of EMIF_CLK clock cycles from Activate (ACTV) to Activate (ACTV) for a different bank, minus 1: $T_RRD = (Trrd/t_{EMIF_CLK}) - 1$
3-0	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.

17.3.7 SDRAM Self Refresh Exit Timing Register (SDSRETR)

The SDRAM self refresh exit timing register (SDSRETR) is used to program the amount of time between when the SDRAM exits Self-Refresh mode and when the EMIF issues another command. The SDSRETR is shown in [Figure 17-21](#) and described in [Table 17-31](#).

Figure 17-21. SDRAM Self Refresh Exit Timing Register (SDSRETR) [offset = 3Ch]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 17-31. SDRAM Self Refresh Exit Timing Register (SDSRETR) Field Descriptions

Bit	Field	Value	Description
31-5	Reserved	0	Reserved. The reserved bit location is always read as 0.
4-0	T_XS	0-1Fh	This field specifies the minimum number of ECLKOUT cycles from Self-Refresh exit to any command, minus 1. $T_XS = T_{Xsr} / t_{EMIF_CLK} - 1$

17.3.8 EMIF Interrupt Raw Register (INTRAW)

The EMIF interrupt raw register (INTRAW) is used to monitor and clear the EMIF's hardware-generated Asynchronous Timeout Interrupt. The AT bit in this register is set when an Asynchronous Timeout occurs regardless of the status of the EMIF interrupt mask set register (INTMSKSET) and EMIF interrupt mask clear register (INTMSKCLR). Writing a 1 to this bit clears it. The EMIF on some devices does not have the EMIF_nWAIT pin; therefore, these registers and fields are reserved on those devices. The INTRAW is shown in [Figure 17-22](#) and described in [Table 17-32](#).

Figure 17-22. EMIF Interrupt Raw Register (INTRAW) [offset = 40h]

31																													8
Reserved																													
R-0																													
7												3	2	1	0														
Reserved												WR		LT		AT													
R-0												R/W1C-0		R/W1C-0		R/W1C-0													

LEGEND: R/W = Read/Write; R = Read only; W1C = Write 1 to clear (writing 0 has no effect); -n = value after reset

Table 17-32. EMIF Interrupt Raw Register (INTRAW) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
2	WR	0 1	Wait Rise. This bit is set to 1 by hardware to indicate that a rising edge on the EMIF_nWAIT pin has occurred. 0 Indicates that a rising edge has not occurred on the EMIF_nWAIT pin. Writing a 0 has no effect. 1 Indicates that a rising edge has occurred on the EMIF_nWAIT pin. Writing a 1 clears this bit and the WR_MASKED bit in the EMIF interrupt masked register (INTMSK).
1	LT	0 1	Line Trap. Set to 1 by hardware to indicate illegal memory access type or invalid cache line size. 0 Writing a 0 has no effect. 1 Indicates that a line trap has occurred. Writing a 1 clears this bit as well as the LT_MASKED bit in the EMIF interrupt masked register (INTMSK).
0	AT	0 1	Asynchronous Timeout. This bit is set to 1 by hardware to indicate that during an extended asynchronous memory access cycle, the EMIF_nWAIT pin did not go inactive within the number of cycles defined by the MAX_EXT_WAIT field in the asynchronous wait cycle configuration register (AWCC). 0 Indicates that an Asynchronous Timeout has not occurred. Writing a 0 has no effect. 1 Indicates that an Asynchronous Timeout has occurred. Writing a 1 clears this bit as well as the AT_MASKED bit in the EMIF interrupt masked register (INTMSK).

17.3.9 EMIF Interrupt Masked Register (INTMSK)

Like the EMIF interrupt raw register (INTRAW), the EMIF interrupt masked register (INTMSK) is used to monitor and clear the status of the EMIF's hardware-generated Asynchronous Timeout Interrupt. The main difference between the two registers is that when the AT_MASKED bit in this register is set, an active-high pulse is sent to the CPU interrupt controller. Also, the AT_MASKED bit field in INTMSK is only set to 1 if the associated interrupt has been enabled in the EMIF interrupt mask set register (INTMSKSET). The EMIF on some devices does not have the EMIF_nWAIT pin, therefore, these registers and fields are reserved on those devices. The INTMSK is shown in [Figure 17-23](#) and described in [Table 17-33](#).

Figure 17-23. EMIF Interrupt Mask Register (INTMSK) [offset = 44h]

31																													8
Reserved																													
R-0																													
7																3	2	1	0										
Reserved															WR_MASKED		LT_MASKED		AT_MASKED										
R-0															R/W1C-0		R/W1C-0		R/W1C-0										

LEGEND: R/W = Read/Write; R = Read only; W1C = Write 1 to clear (writing 0 has no effect); -n = value after reset

Table 17-33. EMIF Interrupt Mask Register (INTMSK) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
2	WR_MASKED	0	Wait Rise Masked. This bit is set to 1 by hardware to indicate a rising edge has occurred on the EMIF_nWAIT pin, provided that the WR_MASK_SET bit is set to 1 in the EMIF interrupt mask set register (INTMSKSET).
		1	Indicates that a wait rise interrupt has not been generated. Writing a 0 has no effect.
		1	Indicates that a wait rise interrupt has been generated. Writing a 1 clears this bit and the WR bit in the EMIF interrupt raw register (INTRAW).
1	LT_MASKED	0	Masked Line Trap. Set to 1 by hardware to indicate illegal memory access type or invalid cache line size, only if the LT_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET) is set to 1.
		1	Writing a 0 has no effect.
		1	Writing a 1 clears this bit as well as the LT bit in the EMIF interrupt raw register (INTRAW).
0	AT_MASKED	0	Asynchronous Timeout Masked. This bit is set to 1 by hardware to indicate that during an extended asynchronous memory access cycle, the EMIF_nWAIT pin did not go inactive within the number of cycles defined by the MAX_EXT_WAIT field in the asynchronous wait cycle configuration register (AWCC), provided that the AT_MASK_SET bit is set to 1 in the EMIF interrupt mask set register (INTMSKSET).
		1	Indicates that an Asynchronous Timeout Interrupt has not been generated. Writing a 0 has no effect.
		1	Indicates that an Asynchronous Timeout Interrupt has been generated. Writing a 1 clears this bit as well as the AT bit in the EMIF interrupt raw register (INTRAW).

17.3.10 EMIF Interrupt Mask Set Register (INTMSKSET)

The EMIF interrupt mask set register (INTMSKSET) is used to enable the Asynchronous Timeout Interrupt. If read as 1, the AT_MASKED bit in the EMIF interrupt masked register (INTMSK) is set and an interrupt is generated when an Asynchronous Timeout occurs. If read as 0, the AT_MASKED bit is always read 0 and no interrupt is generated when an Asynchronous Timeout occurs. Writing a 1 to the AT_MASK_SET bit enables the Asynchronous Timeout Interrupt. The EMIF on some devices does not have the EMIF_nWAIT pin; therefore, these registers and fields are reserved on those devices. The INTMSKSET is shown in [Figure 17-24](#) and described in [Table 17-34](#).

Figure 17-24. EMIF Interrupt Mask Set Register (INTMSKSET) [offset = 48h]

31																	16
Reserved																	
R-0																	
15				3			2			1			0				
Reserved						WR_MASK_SET		Reserved		AT_MASK_SET							
R-0						R/W-0		R-0		R/W-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 17-34. EMIF Interrupt Mask Set Register (INTMSKSET) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
2	WR_MASK_SET	0 1	Wait Rise Mask Set. This bit determines whether or not the wait rise Interrupt is enabled. Writing a 1 to this bit sets this bit, sets the WR_MASK_CLR bit in the EMIF interrupt mask clear register (INTMSKCLR), and enables the wait rise interrupt. To clear this bit, a 1 must be written to the WR_MASK_CLR bit in INTMSKCLR. 0 Indicates that the wait rise interrupt is disabled. Writing a 0 has no effect. 1 Indicates that the wait rise interrupt is enabled. Writing a 1 sets this bit and the WR_MASK_CLR bit in the EMIF interrupt mask clear register (INTMSKCLR).
1	LT_MASK_SET	0 1	Mask set for LT_MASKED bit in the EMIF interrupt mask register (INTMSK). 0 Indicates that the line trap interrupt is disabled. Writing a 0 has no effect. 1 Indicates that the line trap interrupt is enabled. Writing a 1 sets this bit and the LT_MASK_CLR bit in the EMIF interrupt mask clear register (INTMSKCLR).
0	AT_MASK_SET	0 1	Asynchronous Timeout Mask Set. This bit determines whether or not the Asynchronous Timeout Interrupt is enabled. Writing a 1 to this bit sets this bit, sets the AT_MASK_CLR bit in the EMIF interrupt mask clear register (INTMSKCLR), and enables the Asynchronous Timeout Interrupt. To clear this bit, a 1 must be written to the AT_MASK_CLR bit of the EMIF interrupt mask clear register (INTMSKCLR). 0 Indicates that the Asynchronous Timeout Interrupt is disabled. Writing a 0 has no effect. 1 Indicates that the Asynchronous Timeout Interrupt is enabled. Writing a 1 sets this bit and the AT_MASK_CLR bit in the EMIF interrupt mask clear register (INTMSKCLR).

17.3.11 EMIF Interrupt Mask Clear Register (INTMSKCLR)

The EMIF interrupt mask clear register (INTMSKCLR) is used to disable the Asynchronous Timeout Interrupt. If read as 1, the AT_MASKED bit in the EMIF interrupt masked register (INTMSK) is set and an interrupt is generated when an Asynchronous Timeout occurs. If read as 0, the AT_MASKED bit is always read 0 and no interrupt is generated when an Asynchronous Timeout occurs. Writing a 1 to the AT_MASK_CLR bit disables the Asynchronous Timeout Interrupt. The EMIF on some devices does not have the EMIF_nWAIT pin, therefore, these registers and fields are reserved on those devices. The INTMSKCLR is shown in [Figure 17-25](#) and described in [Table 17-35](#).

Figure 17-25. EMIF Interrupt Mask Clear Register (INTMSKCLR) [offset = 4Ch]

31																	16
Reserved																	
R-0																	
15				3			2			1			0				
Reserved				WR_MASK_CLR		Reserved		AT_MASK_CLR									
R-0				R/W-0		R-0		R/W-0									

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 17-35. EMIF Interrupt Mask Clear Register (INTMSKCLR) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reserved. The reserved bit location is always read as 0. If writing to this field, always write the default value of 0.
2	WR_MASK_CLR	0 1	Wait Rise Mask Clear. This bit determines whether or not the wait rise interrupt is enabled. Writing a 1 to this bit clears this bit, clears the WR_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET), and disables the wait rise interrupt. To set this bit, a 1 must be written to the WR_MASK_SET bit in INTMSKSET. 0 Indicates that the wait rise interrupt is disabled. Writing a 0 has no effect. 1 Indicates that the wait rise interrupt is enabled. Writing a 1 clears this bit and the WR_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET).
1	LT_MASK_CLR	0 1	Line trap Mask Clear. This bit determines whether or not the line trap interrupt is enabled. Writing a 1 to this bit clears this bit, clears the LT_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET), and disables the line trap interrupt. To set this bit, a 1 must be written to the LT_MASK_SET bit in INTMSKSET. 0 Indicates that the line trap interrupt is disabled. Writing a 0 has no effect. 1 Indicates that the line trap interrupt is enabled. Writing a 1 clears this bit and the LT_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET).
0	AT_MASK_CLR	0 1	Asynchronous Timeout Mask Clear. This bit determines whether or not the Asynchronous Timeout Interrupt is enabled. Writing a 1 to this bit clears this bit, clears the AT_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET), and disables the Asynchronous Timeout Interrupt. To set this bit, a 1 must be written to the AT_MASK_SET bit of the EMIF interrupt mask set register (INTMSKSET). 0 Indicates that the Asynchronous Timeout Interrupt is disabled. Writing a 0 has no effect. 1 Indicates that the Asynchronous Timeout Interrupt is enabled. Writing a 1 clears this bit and the AT_MASK_SET bit in the EMIF interrupt mask set register (INTMSKSET).

17.3.12 Page Mode Control Register (PMCR)

The page mode control register (PMCR) is shown in [Figure 17-26](#) and described in [Table 17-36](#). This register is configured when using NOR Flash page mode.

Figure 17-26. Page Mode Control Register (PMCR) [offset = 68h]

31	26	25	24
CS5_PG_DEL		CS5_PG_SIZE	CS5_PG_MD_EN
R/W-3Fh		R/W-0	R/W-0
23	18	17	16
CS4_PG_DEL		CS4_PG_SIZE	CS4_PG_MD_EN
R/W-3Fh		R/W-0	R/W-0
15	10	9	8
CS3_PG_DEL		CS3_PG_SIZE	CS3_PG_MD_EN
R/W-3Fh		R/W-0	R/W-0
7	2	1	0
CS2_PG_DEL		CS2_PG_SIZE	CS2_PG_MD_EN
R/W-3Fh		R/W-0	R/W-0

LEGEND: R/W = Read/Write; -n = value after reset

Table 17-36. Page Mode Control Register (PMCR) Field Descriptions

Bit	Field	Value	Description
31-26	CS5_PG_DEL	1-3Fh	Page access delay for NOR Flash connected on CS5. CS5 is not available on this device.
25	CS5_PG_SIZE		Page Size for NOR Flash connected on CS5. CS5 is not available on this device.
24	CS5_PG_MD_EN		Page Mode enable for NOR Flash connected on CS5. CS5 is not available on this device.
23-18	CS4_PG_DEL	1-3Fh	Page access delay for NOR Flash connected on CS4. Number of EMIF_CLK cycles required for the page read data to be valid, minus 1 cycle. This value must not be cleared to 0.
17	CS4_PG_SIZE		Page Size for NOR Flash connected on CS4.
		0	Page size is 4 words.
		1	Page size is 8 words.
16	CS4_PG_MD_EN		Page Mode enable for NOR Flash connected on CS4.
		0	Page mode is disabled for this chip select.
		1	Page mode is enabled for this chip select.
15-10	CS3_PG_DEL	1-3Fh	Page access delay for NOR Flash connected on CS3. Number of EMIF_CLK cycles required for the page read data to be valid, minus 1 cycle. This value must not be cleared to 0.
9	CS3_PG_SIZE		Page Size for NOR Flash connected on CS3.
		0	Page size is 4 words.
		1	Page size is 8 words.
8	CS3_PG_MD_EN		Page Mode enable for NOR Flash connected on CS3.
		0	Page mode is disabled for this chip select.
		1	Page mode is enabled for this chip select.
7-2	CS2_PG_DEL	1-3Fh	Page access delay for NOR Flash connected on CS2. Number of EMIF_CLK cycles required for the page read data to be valid, minus 1 cycle. This value must not be cleared to 0.
1	CS2_PG_SIZE		Page Size for NOR Flash connected on CS2.
		0	Page size is 4 words.
		1	Page size is 8 words.
0	CS2_PG_MD_EN		Page Mode enable for NOR Flash connected on CS2.
		0	Page mode is disabled for this chip select.
		1	Page mode is enabled for this chip select.

17.4 Example Configuration

This section presents an example of interfacing the EMIF to both an SDR SDRAM device and an asynchronous flash device.

17.4.1 Hardware Interface

Figure 17-27 shows the hardware interface between the EMIF, a Samsung K4S641632H-TC(L)70 64Mb SDRAM device, and two SHARP LH28F800BJE-PTTL90 8Mb Flash memory. The connection between the EMIF and the SDRAM is straightforward, but the connection between the EMIF and the flash deserves a detailed look.

The address inputs for the flash are provided by three sources. The A[18:0] address inputs are provided by a combination of the EMIF_A and EMIF_BA pins according to Section 17.2.6.1. RD/nBY signal from one flash is connected to EMIF_nWAIT pin of EMIF.

Finally, this example configuration connects the EMIF_nWE pin to the nWE input of the flash and operates the EMIF in Select Strobe Mode.

17.4.2 Software Configuration

The following sections describe how to configure the EMIF registers and bit fields to interface the EMIF with the Samsung K4S641632H-TC(L)70 SDRAM and the SHARP LH28F800BJE-PTTL90 8Mb Flash memory.

17.4.2.1 Configuring the SDRAM Interface

This section describes how to configure the EMIF to interface with the Samsung K4S641632H-TC(L)70 SDRAM with a clock frequency of $f_{\text{EMIF_CLK}} = 100 \text{ MHz}$. Procedure A described in Section 17.2.5.5 is followed which assumes that the SDRAM power-up timing constraint were met during the SDRAM Auto-Initialization sequence after Reset.

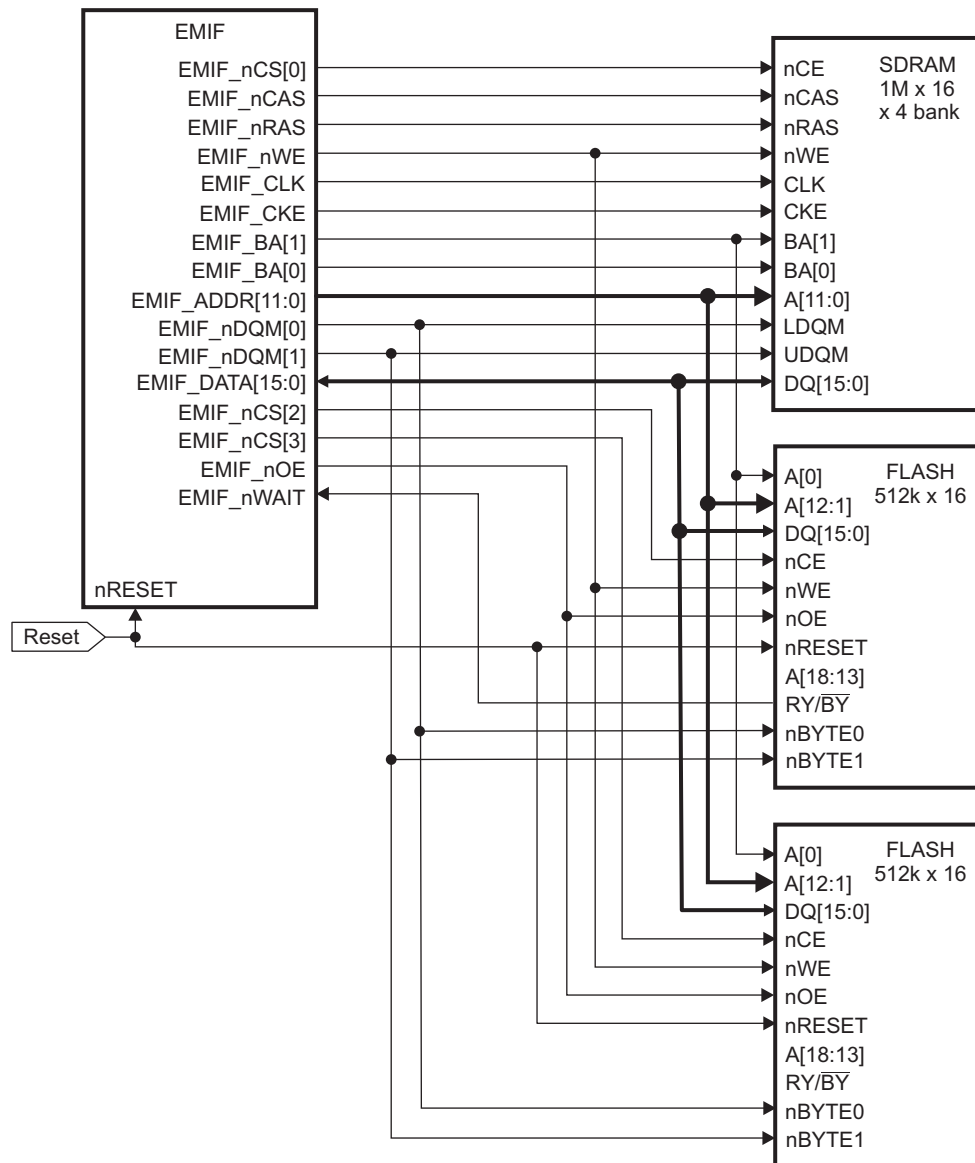
17.4.2.1.1 PLL Programming for the EMIF to K4S641632H-TC(L)70 Interface

The device global clock module (GCM) should first be programmed to select the desired EMIF_CLK frequency. Before doing this, the SDRAM should be placed in Self-Refresh Mode by setting the SR bit in the SDRAM configuration register (SDCR). The SR bit should be set using a byte-write to the upper byte of the SDCR to avoid triggering the SDRAM Initialization Sequence. The EMIF_CLK frequency can now be configured to the desired value by selecting the appropriate clock source for the VCLK3 domain. Once the VCLK3 domain frequency has been configured, remove the SDRAM from Self-Refresh by clearing the SR bit in SDCR, again with a byte-write.

Table 17-37. SR Field Value For the EMIF to K4S641632H-TC(L)70 Interface

Field	Value	Purpose
SR	1 then 0	To place the EMIF into the self refresh state

Figure 17-27. Example Configuration Interface



17.4.2.1.2 SDRAM Timing Register (SDTIMR) Settings for the EMIF to K4S641632H-TC(L)70 Interface

The fields of the SDRAM timing register (SDTIMR) should be programmed first as described in [Table 17-38](#) to satisfy the required timing parameters for the K4S641632H-TC(L)70. Based on these calculations, a value of 6111 4610h should be written to SDTIMR. [Figure 17-28](#) shows a graphical description of how SDTIMR should be programmed.

Table 17-38. SDTIMR Field Calculations for the EMIF to K4S641632H-TC(L)70 Interface

Field Name	Formula	Value from K4S641632H-TC(L)70 Datasheet	Value Calculated for Field
T_RFC	$T_RFC \geq (t_{RFC} \times f_{EMIF_CLK}) - 1$	$t_{RC} = 68 \text{ ns (min)}^{(1)}$	6
T_RP	$T_RP \geq (t_{RP} \times f_{EMIF_CLK}) - 1$	$t_{RP} = 20 \text{ ns (min)}$	1
T_RCD	$T_RCD \geq (t_{RCD} \times f_{EMIF_CLK}) - 1$	$t_{RCD} = 20 \text{ ns (min)}$	1
T_WR	$T_WR \geq (t_{WR} \times f_{EMIF_CLK}) - 1$	$t_{RDL} = 2 \text{ CLK} = 20 \text{ ns (min)}^{(2)}$	1
T_RAS	$T_RAS \geq (t_{RAS} \times f_{EMIF_CLK}) - 1$	$t_{RAS} = 49 \text{ ns (min)}$	4
T_RC	$T_RC \geq (t_{RC} \times f_{EMIF_CLK}) - 1$	$t_{RC} = 68 \text{ ns (min)}$	6
T_RRD	$T_RRD \geq (t_{RRD} \times f_{EMIF_CLK}) - 1$	$t_{RRD} = 14 \text{ ns (min)}$	1

⁽¹⁾ The Samsung datasheet does not specify a t_{RFC} value. Instead, Samsung specifies t_{RC} as the minimum auto refresh period.

⁽²⁾ The Samsung datasheet does not specify a t_{WR} value. Instead, Samsung specifies t_{RDL} as last data in to row precharge minimum delay.

Figure 17-28. SDRAM Timing Register (SDTIMR)

31	27	26	24	23	22	20	19	18	16
0 0110	001	0	001	0	001	0	001		
T_RFC	T_RP	Rsvd	T_RCD	Rsvd	T_WR				
15	12	11	8	7	6	4	3		0
0100	0110	0	001	0000					
T_RAS	T_RC	Rsvd	T_RRD	Reserved					

17.4.2.1.3 SDRAM Self Refresh Exit Timing Register (SDSRETR) Settings for the EMIF to K4S641632H-TC(L)70 Interface

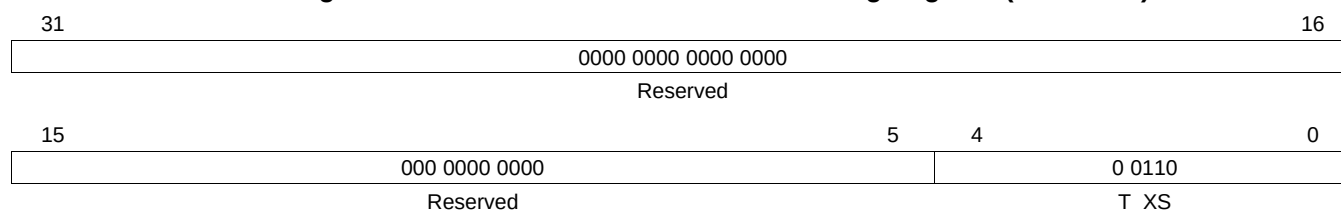
The SDRAM self refresh exit timing register (SDSRETR) should be programmed second to satisfy the t_{XSR} timing requirement from the K4S641632H-TC(L)70 datasheet. [Table 17-39](#) shows the calculation of the proper value to program into the T_XS field of this register. Based on this calculation, a value of 6h should be written to SDSRETR. [Figure 17-29](#) shows how SDSRETR should be programmed.

Table 17-39. RR Calculation for the EMIF to K4S641632H-TC(L)70 Interface

Field Name	Formula	Value from K4S641632H-TC(L)70 Datasheet	Value Calculated for Field
T_XS	$T_XS \geq (t_{XSR} \times f_{EMIF_CLK}) - 1$	$t_{RC} = 68 \text{ ns (min)}$ ⁽¹⁾	6

⁽¹⁾ The Samsung datasheet does not specify a t_{XSR} value. Instead, Samsung specifies t_{RC} as the minimum required time after CKE going high to complete self refresh exit.

Figure 17-29. SDRAM Self Refresh Exit Timing Register (SDSRETR)



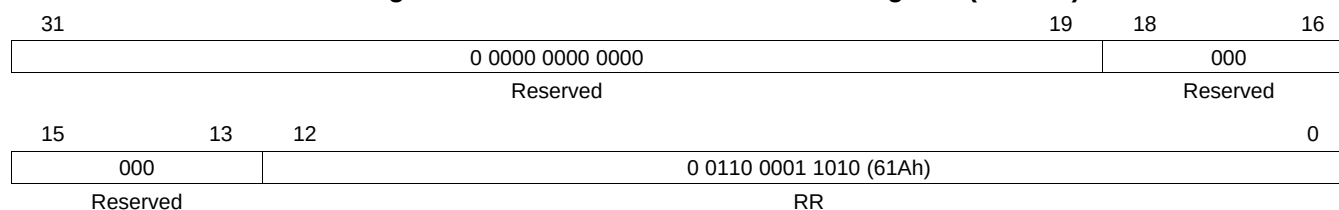
17.4.2.1.4 SDRAM Refresh Control Register (SDRCR) Settings for the EMIF to K4S641632H-TC(L)70 Interface

The SDRAM refresh control register (SDRCR) should next be programmed to satisfy the required refresh rate of the K4S641632H-TC(L)70. [Table 17-40](#) shows the calculation of the proper value to program into the RR field of this register. Based on this calculation, a value of 61Ah should be written to SDRCR. [Figure 17-30](#) shows how SDRCR should be programmed.

Table 17-40. RR Calculation for the EMIF to K4S641632H-TC(L)70 Interface

Field Name	Formula	Values	Value Calculated for Field
RR	$RR \leq f_{EMIF_CLK} \times t_{Refresh\ Period} / n_{cycles}$	From SDRAM datasheet: $t_{Refresh\ Period} = 64\ ms$; $n_{cycles} = 4096$ EMIF clock rate: $f_{EMIF_CLK} = 100\ MHz$	RR = 1562 cycles = 61Ah cycles

Figure 17-30. SDRAM Refresh Control Register (SDRCR)



17.4.2.1.5 SDRAM Configuration Register (SDCR) Settings for the EMIF to K4S641632H-TC(L)70 Interface

Finally, the fields of the SDRAM configuration register (SDCR) should be programmed as described in [Table 17-37](#) to properly interface with the K4S641632H-TC(L)70 device. Based on these settings, a value of 4720h should be written to SDCR. [Figure 17-31](#) shows how SDCR should be programmed. The EMIF is now ready to perform read and write accesses to the SDRAM.

Table 17-41. SDCR Field Values For the EMIF to K4S641632H-TC(L)70 Interface

Field	Value	Purpose
SR	0	To avoid placing the EMIF into the self refresh state
NM	1	To configure the EMIF for a 16-bit data bus
CL	011b	To select a CAS latency of 3
BIT11_9LOCK	1	To allow the CL field to be written
IBANK	010b	To select 4 internal SDRAM banks
PAGESIZE	0	To select a page size of 256 words

Figure 17-31. SDRAM Configuration Register (SDCR)

31	30	29	28	24
0	0	0	0 0000	
SR	Reserved	Reserved	Reserved	
23			18	17
		00 0000	0	0
		Reserved	Reserved	Reserved
15	14	13	12	11
0	1	0	0	011
Reserved	NM	Reserved	Reserved	CL
				BIT11_9LOCK
7	6		4	3
0	010		0	000
Reserved	IBANK		Reserved	PAGESIZE

17.4.2.2 Configuring the Flash Interface

This section describes how to configure the EMIF to interface with the two of SHARP LH28F800BJE-PTTL90 8Mb Flash memory with a clock frequency of $f_{\text{EMIF_CLK}} = 100 \text{ MHz}$. The example assumes that one flash is connected to EMIF_nCS2 and the other to EMIF_nCS3.

17.4.2.2.1 Asynchronous 1 Configuration Register (CE2CFG) Settings for the EMIF to LH28F800BJE-PTTL90 Interface

The asynchronous 1 configuration register (CE2CFG) and asynchronous 2 configuration register (CE3CFG) are the only registers that is necessary to program for this asynchronous interface (assuming that one Flash is connected to EMIF_nCS[2] and the other to EMIF_nCS[3]. The SS bit (in both registers) should be set to 1 to enable Select Strobe Mode and the ASIZE field (in both registers) should be set to 1 to select a 16-bit interface. The other fields in this register control the shaping of the EMIF signals, and the proper values can be determined by referring to the AC Characteristics in the Flash datasheet and the device datasheet. Based on the following calculations, a value of 8862 25BDh should be written to CE2CFG. [Table 17-42](#) and [Table 17-43](#) show the pertinent AC Characteristics for reads and writes to the Flash device, and [Figure 17-32](#) and [Figure 17-33](#) show the associated timing waveforms. Finally, [Figure 17-34](#) shows programming the CE n CFG ($n = 2, 3$) with the calculated values.

Table 17-42. AC Characteristics for a Read Access

AC Characteristic	Device	Definition	Min	Max	Unit
t_{SU}	EMIF	Setup time, read EMIF_D before EMIF_CLK high	6.5		ns
t_{H}	EMIF	Data hold time, read EMIF_D after EMIF_CLK high	1		ns
t_{D}	EMIF	Output delay time, EMIF_CLK high to output signal valid		7	ns
t_{ELQV}	Flash	nCE to Output Delay		90	ns
t_{EHQZ}	Flash	nCE High to Output in High Impedance		55	ns

Table 17-43. AC Characteristics for a Write Access

AC Characteristic	Device	Definition	Min	Max	Unit
t_{AVAV}	Flash	Write Cycle Time	90		ns
t_{ELEH}	Flash	nCE Pulse Width Low	50		ns
t_{EHEL}	Flash	nCE Pulse Width High (not shown in Figure 17-33)	30		ns

Figure 17-32. LH28F800BJE-PTTL90 to EMIF Read Timing Waveforms

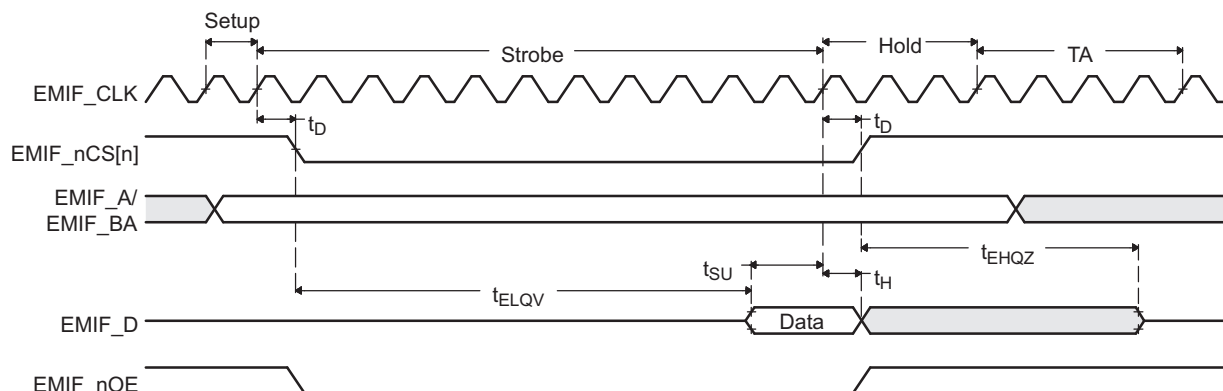
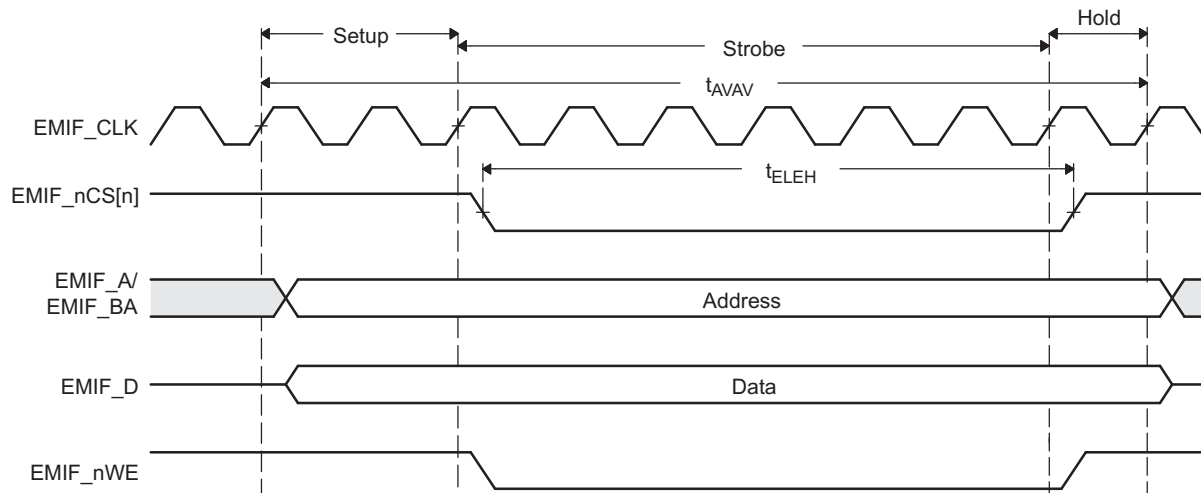


Figure 17-33. LH28F800BJE-PTTL90 to EMIF Write Timing Waveforms


The R_STROBE field should be set to meet the following equation:

$$R_STROBE \geq (t_D + t_{ELQV} + t_{SU}) \times f_{EMIF_CLK} - 1$$

$$R_STROBE \geq (7 \text{ ns} + 90 \text{ ns} + 6.5 \text{ ns}) \times 100 \text{ MHz} - 1$$

$$R_STROBE \geq 9.35$$

$$R_STROBE = 10$$

The R_HOLD field must be large enough to satisfy the EMIF Data hold time, t_H :

$$R_HOLD \geq t_H \times f_{EMIF_CLK} - 1$$

$$R_HOLD \geq 1 \text{ ns} \times 100 \text{ MHz} - 1$$

$$R_HOLD \geq -0.9$$

The R_HOLD field must also combine with the TA field to satisfy the Flash's nCE High to Output in High Impedance time, t_{EHQZ} :

$$R_HOLD + TA \geq (t_D + t_{EHQZ}) \times f_{EMIF_CLK} - 2$$

$$R_HOLD + TA \geq (7 \text{ ns} + 55 \text{ ns}) \times 100 \text{ MHz} - 2$$

$$R_HOLD + TA \geq 4.2$$

The largest value that can be programmed into the TA field is 3h, therefore the following values can be used:

$$R_HOLD = 2$$

$$TA = 3$$

For Writes, the W_STROBE field should be set to satisfy the Flash's nCE Pulse Width constraint, t_{ELEH} :

$$W_STROBE \geq t_{ELEH} \times f_{EMIF_CLK} - 1$$

$$W_STROBE \geq 50 \text{ ns} \times 100 \text{ MHz} - 1$$

$$W_STROBE \geq 4$$

The W_SETUP and W_HOLD fields should combine to satisfy the Flash's nCE Pulse Width High constraint, t_{EHEL} , when performing back-to-back writes:

$$W_SETUP + W_HOLD \geq t_{EHEL} \times f_{EMIF_CLK} - 2$$

$$W_SETUP + W_HOLD \geq 30 \text{ ns} \times 100 \text{ MHz} - 2$$

$$W_SETUP + W_HOLD \geq 1$$

In addition, the entire Write access length must satisfy the Flash's minimum Write Cycle Time, t_{AVAV} :

$$W_SETUP + W_STROBE + W_HOLD \geq t_{AVAV} \times f_{EMIF_CLK} - 3$$

$$W_SETUP + W_STROBE + W_HOLD \geq 90 \text{ ns} \times 100 \text{ MHz} - 3$$

$$W_SETUP + W_STROBE + W_HOLD \geq 6$$

Solving the above equations for the Write fields results in the following possible solution:

$$W_SETUP = 1$$

$$W_STROBE = 5$$

$$W_HOLD = 0$$

Adding a 10 ns (1 cycle) margin to each of the periods (excluding TA which is already at its maximum) in this example produces the following recommended values:

$$W_SETUP = 2h$$

$$W_STROBE = 6h$$

$$W_HOLD = 1h$$

$$R_SETUP = 1h$$

$$R_STROBE = 8h$$

$$R_HOLD = 3h$$

$$TA = 3h$$

Figure 17-34. Asynchronous m Configuration Register ($m = 1, 2$) (CE $_n$ CFG ($n = 2, 3$))

31	30	29	26	25	24
1	0	0010	00		
SS	EW	W_SETUP	W_STROBE		
23	20	19	17	16	
0110	001	0			
W_STROBE	W_HOLD	R_SETUP			
15	13	12	7	6	4
001	001011	011	11	01	
R_SETUP	R_STROBE	R_HOLD	TA	ASIZE	